Hints Dilute and Rotate the Evidence Axis of Vision-Language Models

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Abstract

Vision-language models often answer when the image lacks the queried evidence, yet retract a correct answer as soon as a prompt sentence contradicts it. Are these failures of seeing or of reporting? We use a 2×3 within-subject design—visual evidence {present, annotation-absent} $\times$ textual hint {none, correct, conflicting} (the absent arm has no correct colour, so its middle cell asserts a plausible one)—built without editing a pixel: "absent" is a category that COCO annotations place outside the photograph, checked by a presence probe. Across 261 images and three VLMs built on Qwen2/2.5 language models, conflict collapses the models' own output signals: $p(\texttt{unknown})$ falls from AUROC 0.993 to 0.834 in Qwen2.5-VL-3B, and maximum probability reaches chance (0.463). A probe fitted only on the no-hint pair keeps its ranking on held-out scenes (0.96, 0.94, 0.92 in 3B, 7B and LLaVA, plausible hint), so the evidence is still there; what changes is its geometry. The full present–absent mean-difference vector shortens to 0.39–0.64 of its no-hint length and turns 24°–71° away from the no-hint axis, with the two hinted colours rotating to nearly the same direction ($\cos = 0.91$); per-scene gaps lose coherence ($0.70 \rightarrow 0.41$ in 7B), while noise-normalised separation does not fall and in-condition refitting recovers at least 0.989; because the within-class spread contracts by a nearly constant 0.71, the fixed readout's rank-sensitivity loss is proportional to the mean-gap contraction (d' ratio = 1.42$\times$ gap ratio, $R^2 = 0.92$ across four models, leave-one-cell-out error 0.06). Two further backbones outside that family repeat the geometry: Gemma-3-4B (Gemma-3 with a SigLIP encoder) at 0.53 of the length, 24°–29° of rotation, $\cos = 0.97$ and 0.96 held-out transfer, and Phi-3.5-vision (Phi-3 with a CLIP encoder) at 0.05–0.15 of the length and 74°–86° of rotation. Gemma's output signal barely degrades while Phi's collapses ($p(\texttt{unknown})$ $0.939 \rightarrow 0.385$), so the geometry generalises across three language-model families, while the collapse tracks the readout rather than the family. The hint does not erase the evidence; it moves it off the axis the readout was using, and dilutes it. Signal detection separates dilution from criterion shift: sensitivity falls in all three Qwen-family models (d' $3.51 \rightarrow 1.45$, $2.98 \rightarrow 1.24$, $3.44 \rightarrow 1.44$), criterion shifts are inconsistent, most of the flattened curve is the axis. Pasting the no-hint state back at 69–94% depth returns 80–102% of the lost refusal in all five models, so the gate is written late. Steering shows readout-level sufficiency; energy-matched removal shows no direction-specific necessity, so we claim readout-level sufficiency, not a circuit. A frozen gate cuts conflict fabrication at 25% answered coverage from 0.192 to 0.077 (3B), 0.169 to 0.146 (7B), 0.138 to 0.062 (LLaVA), 0.138 to 0.015 (Gemma) and 0.562 to 0.285 (Phi), but a category-matched control that holds the question fixed across the label removes the 7B and LLaVA margin, so part of the gain is arm-specific; conformal certification of a fabrication rate is feasible only in a minority of splits at this sample size, and we report the minimum bound instead; on naturally unanswerable VizWiz questions $p(\texttt{unknown})$ matches the probe. Code is released.

1 Introduction

Ask a vision-language model (VLM) *"What color is the umbrella?"* about a photograph with no umbrella, and it will usually answer `unknown`. Prepend one plausible sentence—*"Hint: the umbrella is black."*—and the same model answers *black* with high confidence. The failure is not that the model cannot see. The object was never there.

Two literatures describe adjacent phenomena. Work on hallucination measures objects a model reports but the image does not contain [19, 24]. Work on sycophancy measures answers that follow a user's stated belief rather than the input [13, 25, 28]. Both describe behaviour; neither separates two questions that the behaviour confounds: *is the evidence represented?* and *is the decision to abstain driven by that representation?* A model that answers *black* may have lost the evidence, or it may still represent "no umbrella" while a later stage overrides abstention.

We separate the two with a design in which the image is held fixed and only the *epistemic status* of the question changes. Each of 261 photographs appears in six cells: the queried object is present or annotated-absent, crossed with no hint, a correct hint, or a conflicting hint. The absent arm requires no image editing: it asks about a category that COCO's instance annotations place outside the image. Every cell shares the same pixels, and the six prompts differ only in what they assert. A separate yes/no probe assigns mean $p(\text{yes}) = 0.095$ to the absent images versus 0.95 to the originals; annotation absence is independently checked, but it is not identical to model-perceived absence.

The design yields a dissociation (Figure 1). With no hint, the model's own $p(\texttt{unknown})$ detects evidence availability almost perfectly (AUROC 0.993). Under conflict every output-space signal degrades ($p(\texttt{unknown})$ 0.834, maximum probability 0.463, 12-way readout 0.923), while a probe fitted only on the no-hint pair keeps its ranking on scenes it never saw (0.995 refitted, 0.958 held out, hint `b`) yet its present–absent geometry changes sharply. Measuring the full mean-difference vector, rather than its projection onto one direction, separates the ways a separation can shrink. Under a conflicting hint the vector shortens to 0.53 of its no-hint length and turns 40° away from the no-hint axis in 3B; the same rotation appears in 7B and LLaVA, in a non-Qwen backbone (24°–29° in Gemma-3-4B) and reaches 86° in a second one (Phi-3.5-vision), and two different hinted colours rotate to almost the same direction ($\cos = 0.91$), so it is systematic rather than colour noise. The along-axis projection, which is what a fixed readout sees, falls further than either factor alone (0.41 in 3B), while noise-normalised separation does not fall and a readout refitted inside the condition recovers 0.995–0.999. The hint dilutes and rotates the evidence axis; it does not erase it. Three tests follow. Signal detection shows a sensitivity loss with inconsistent criterion shifts, most of the flattened refusal curve being predicted by the dilution. Patching restores most of the lost abstention only in the final few layers, and a direction injected at the readout controls abstention bidirectionally; we claim sufficiency at the readout, not an identified circuit, because removal at matched energy shows no direction-specific necessity. A frozen gate trained only on the no-hint pair then cuts fabrication under conflict by 2.2–2.5× in 3B and LLaVA at a quarter of the items answered, a gain that survives prevalence shift but not the move to naturally unanswerable images; in 7B the probe is nominally ahead of $p(\texttt{unknown})$ but the difference is within noise.

Contributions.

- **A controlled construction.** A 2×3 within-subject benchmark with no image editing (not fully crossed: absent items have no correct colour, so the middle hint cell is a plausible conflicting one), in which "evidence absent" is annotation-absent and independently checked by a presence probe, and in which the conflicting hint is plausibility-matched to the present arm. The 261-item manifest and labels are frozen and released.
- **A rank–scale dissociation with a bounded causal claim.** Native output signals collapse under conflict while the internal evidence representation stays rank-decodable on held-out scenes but dilutes and rotates. We measure the full mean-difference geometry rather than a single projection, report scene-held-out transfer rather than refitted performance as the headline, and quantify the link between geometry and rank sensitivity: the fixed readout's d' loss is proportional to the mean-gap contraction, with a per-model gain of 1.25–1.60 ($R^2 = 0.918$ over 12 model–condition cells). We downgrade the causal claim to readout-level sufficiency after an energy-matched removal audit; and a behavioural check shows the dilution law transfers to the model's own output signal in direction and order of magnitude, with a model-specific gain.
- **A deployment result and its boundary.** A scene-cross-fitted linear gate reduces fabrication by 2.2–2.5× in 3B and LLaVA, 9.0× in Gemma and 2.0× in Phi at 25% answered coverage, survives prevalence shifts from 5% to 50%, and loses its advantage on naturally unanswerable images; in 7B the nominal 1.2× difference has a scene-bootstrap interval of $[-0.062, +0.008]$ and is not distinguishable from zero. A category-matched control, in which both arms ask about the same category, keeps the geometry and the in-condition refit but removes the 7B and LLaVA margin over $p(\texttt{unknown})$, reverses Gemma's and leaves Phi's at 0.008, so the gain is partly specific to the main design's change of question; we report both. We test conformal certification explicitly and report that 130 calibration scenes support a family-wise 10% certificate in only 4/20 splits, and in no split for the other models, under a consistent risk denominator; the deployment claim is empirical, not distribution-free.

2 Related Work

Sycophancy and prompt-induced error. Language models shift answers toward a user's stated view [25], and their explanations can diverge from the evidence that drove the answer [28]. Recent work documents the multimodal version: VLMs change visual answers to agree with textual assertions, sometimes while still encoding the original content [13]; the phenomenon is also described as blind faith in text [8]. These studies are behavioural or input-level. We intervene on hidden states, decompose the decision into sensitivity and criterion, and use the resulting direction to build a gate.

Abstention and selective prediction. Selective prediction trades coverage for risk using a confidence score [10], and has been studied for VQA under distribution shift and on unanswerable questions [29]. ReCoVERR reduces over-abstention by acquiring more evidence [26], black-box consistency identifies unreliable answers [16], and Rep2Act trains a representation-to-action router that turns latent answerability into an explicit abstention action [12]. These methods change the decision procedure or the training objective. We hold both fixed and ask *which representation* the scalar should be computed from, and show that the model's own abstention probability is the wrong representation when a hint is present.

Distribution-free risk control. Conformal prediction and conformal risk control give finite-sample guarantees for a chosen loss under exchangeability [2, 6]. We apply split conformal risk control with a binomial upper confidence bound to fabrication risk and report that, with 130 calibration scenes and a consistent answered-set denominator, no family-wise 5% or 10% certificate is feasible; the minimum bound is reported instead. BCEA couples conformal filtering with evidence acquisition [30], and A^2Safe grounds VQA safety decisions in a multi-agent evidence board [31]; we test whether a fabrication rate can be certified under prompt pollution without new evidence or an agent layer.

Probing and steering internal states. Linear probes recover truthfulness, knowledge and uncertainty from hidden states [1, 4, 7, 15, 21]. Difference-of-means directions can steer behaviour [18, 23, 27, 33], and chat-model refusal is mediated by a low-dimensional direction [3, 14]. Activation steering has been applied to VLM refusal [32]. We study a different axis—whether the evidence needed to answer is available—and pair steering with a projection audit that bounds what the ablation can show.

Localisation and hallucination. Activation patching replaces a state in one forward pass with a state from another and measures the effect on the output [22]. Object hallucination is typically measured by asking about absent objects [19, 24]. We adopt the logic of absent-object questions but make the comparison within-image and within-question, vary only the textual assertion, and add an internal readout.

3 Design and Data

3.1 The 2×3 design

Each sample is a photograph, a queried category c, and a gold colour g. The six conditions are:

	textual hint		
visual evidence	none	correct	conflicting
object present	`sufficient|none`	`sufficient|correct`	`sufficient|wrong`
object absent	`absent|none`	`absent|a`	`absent|b`

In the present arm the correct hint states g and the conflicting hint states a fixed far-hue distractor. In the absent arm there is no correct colour: `a` is the absent category's most frequent colour in COCO val (plausible for the category), and `b` applies the same far-hue mapping to that colour. `b` is therefore plausibility-matched to `sufficient|wrong`, which also uses a far hue; we use `b` for the geometry contrast and report `a` separately throughout. The plausible hint `a` is the harder deployment case (it induces more credulity) and is the one used for the gate. Every image appears in all six cells, so image identity is held constant; within an arm the queried

category is fixed as well. The present and absent arms query different categories by construction, so present-versus-absent contrasts are within-image but not within-question. Appendix D reports a category-matched control that holds the queried category and the asserted colour fixed and varies only the evidence.

3.2 Construction without image editing

Starting from COCO val2017 [20], we retain single annotated instances with box short side $\geq$ 44 px, area fraction 1.5–35%, and mask/box fill ≥ 0.45. Colour is proposed as the area-dominant colour inside the instance mask, with purity and margin gates, then verified by hand on contact sheets. Of 1447 geometric candidates, 413 passed the automatic colour gate and 261 survived manual verification (blue 88, white 79, black 45, green 25, red 15, yellow 9; Appendix C).

For the absent arm we sample a category with no annotation in the image (74 distinct categories appear) and use that category's {first, second} most frequent colour in val as the hint. The colour is plausible for the category; we do not constrain the category to be plausible for the scene. No pixel is edited; the same JPEG is used in all six cells.

Manipulation check. A separate yes/no question ("Is there a c in this image?") yields mean $p(\text{yes}) = 0.095$ on 260 category-matched negative images versus 0.95 on the originals. The absent arm is therefore annotation-absent and independently checked. Because 6.5% of these negatives are judged present with probability above 0.5, we report a strict-absent sensitivity analysis in Appendix D and treat "absent" as annotation-level throughout.

Absence is also not the only difference between the arms: a category that fits the scene is refused less often than one that does not. Scoring each absent category by its COCO train co-occurrence with the objects annotated in the image and splitting the items at the median, no-hint refusal is lower in the better-fitting half by 0.116 $[0.047, 0.185]$ (3B), 0.069 $[0.023, 0.115]$ (7B) and 0.132 $[0.039, 0.224]$ (LLaVA). The evidence axis is unaffected: within each half, $p(\texttt{unknown})$ still separates present from absent images at AUROC ≥ 0.976 and the internal probe at ≥ 0.989 (Appendix D). Scene fit moves the level of refusal, not the ordering of present against absent. The effect survives a control for category frequency, which is only weakly related to it.

3.3 Prompts, readout and signals

The system prompt asks for one word and instructs the model to answer `unknown` when the image does not contain enough information. The user prompt is

```
[Hint:  the {category} is {colour}.]
What color is the {category}?
Answer with one word.
```

where the hint line is omitted in the no-hint condition. The readout is a distribution over eleven colour words plus `unknown`; all twelve candidates are single tokens in every tokenizer used. Probabilities are the softmax over their logits at the first answer position.

Output signals. We compute $p(\texttt{unknown})$, maximum probability over the twelve candidates, the top-two gap, negative entropy, and a 12-dimensional logistic readout over the full candidate distribution. These use the same forward pass as the answer.

Internal probe. A logistic regression on the final-token hidden state (2048-d for 3B, 3584-d for the two 7B models), with standardised features and $C = 0.1$. It is trained on the no-hint pair `sufficient|none` vs. `absent|none`. Cross-validation is grouped by scene (five folds) so a scene never appears in both train and test; a second analysis groups by queried category to test a category→colour shortcut. The *frozen* protocol fits on the no-hint states of the training scenes and scores the *held-out* scenes' hinted states, so no scene contributes both to fitting and to evaluation; the headline numbers use this protocol at a fixed $C = 0.1$. The *refit* protocol retrains inside each condition, still grouped by scene, and is reported only as an upper bound. The direction used for steering uses $C = 0.01$ because it is injected at a fixed norm; Appendix A records both settings and the sensitivity to C.

Table 1: Behaviour on 261 verified samples (Qwen2.5-VL-3B). "Adopts hint" is the fraction whose argmax equals the hinted colour. On absent rows, 1 − refusal is the unanswerable answer rate (UAR); on present rows, refusal is the over-refusal rate (OAR). Cross-model refusal is reported in the text.

condition	refusal	mean conf.	adopts hint	accuracy
present · correct hint	0.004	0.943	—	0.973
present · conflicting hint	0.088	0.674	0.448	0.383
present · no hint	0.011	0.836	—	0.762
absent · hint a (plausible)	0.521	0.699	0.467	—
absent · hint b (far hue)	0.563	0.707	0.433	—
absent · no hint	0.904	0.799	—	—

Steering and patching. The direction w is the logistic-regression probe direction fitted on the no-hint pair. We inject $\alpha\|h\|w$ at the final position, where $\|h\|$ is the median hidden-state norm and $\alpha \in \{\pm 0.25, \pm 0.5\}$, and ablate by projecting w out of the state. A split-half check fits w on 130 scenes and evaluates on the remaining 131. For patching, layer L receives the final-token state from a no-hint pass and the remaining layers run; a shuffled-source control pastes another sample's no-hint state at the same layer.

Metrics. Evidence detection is AUROC over the 522 paired items, with scene-grouped bootstrap confidence intervals (2000 resamples) and paired bootstrap intervals for differences. Selective answering is reported as the fabrication rate among answered items at matched answered coverage; 25% is the shared operating point (feasible for every model), and 50% is reported where the model's own answering rate allows it. We report unadjusted 95% intervals and treat them as descriptive: the primary contrasts exclude zero by wide margins, while the own-axis residual reported below is marginal and is labelled as such.

Models. Qwen2.5-VL-3B and 7B [5] in bfloat16 with SDPA attention and at most 1024^2 pixels, and LLaVA-OneVision-7B [17] with the longest image side at most 512 px, loaded through `AutoModelForImageTextToText`. All LLaVA stages use the same 512 px preprocessing. All experiments ran on a single 24 GB A10.

4 Results

4.1 Behaviour: a hint moves both arms in opposite directions

Table 1 shows the basic pattern. Without hints the models are conservative in the right way: 3B refuses 90.4% of absent-object questions and 1.1% of present-object ones. A conflicting hint moves both arms toward the assertion: refusal on the absent arm falls to 52.1% (hint a) and 56.3% (hint b), and 43–47% of answers name the hinted colour. On the present arm the same hint raises refusal from 1.1% to 8.8% and drops accuracy from 76.2% to 38.3%. A correct hint does the opposite, raising accuracy to 97.3%. The hint is therefore treated as evidence, not noise: it helps when true and hurts when false. The two absent hints differ in effect size but not in sign, and the far-hue hint `b` is the closer match to the present-arm manipulation.

The base policies differ sharply. Qwen2.5-VL-7B over-refuses under conflict (55.6% on present items) and adopts the hint on only 17.6%; LLaVA-OneVision is close to fully credulous (present refusal 0.4%, absent-hint adoption 76–90%). Gemma-3-4B-it—Gemma-3 with a SigLIP encoder, one of two backbones here whose language model is not from the Qwen family (the other is Phi-3.5-vision, Appendix G)—sits between them: it refuses 84.3% of absent items without a hint, conflict lowers this to 40.2% (hint a) and 47.5% (hint b) with adoption of 57.9% and 50.2%, and on present items the same hint is adopted on 67.4% and drops accuracy from 56.7% to 24.1%. Its colour reading without a hint is weaker than 3B's, so the present arm is noisier, but the direction of both arms is the same. All five show the dilution and rotation documented next (Phi-3.5 in Appendix G), which is why we treat the geometry as the invariant and the behavioural form as model-specific.

4.2 Native confidence collapses; the evidence axis dilutes and rotates

Table 2 and Figure 1 contain the central result. With no hint, $p(\texttt{unknown})$ is already an excellent evidence detector (0.993), statistically indistinguishable from the probe ($+0.002$ $[-0.003, +0.007]$). The interesting regime is conflict. There $p(\texttt{unknown})$ loses 0.159 AUROC, maximum probability loses 0.152 and falls to chance, and even a learned readout over the entire output distribution loses 0.059. The frozen probe loses 0.038 and transfers

Table 2: Evidence detection (AUROC over 522 paired items, Qwen2.5-VL-3B) from the same forward pass. The conflict pair states a far-hue colour on present items and the plausible colour on absent items; the geometry contrast below uses the far-hue hint on both arms. "Frozen" fits the probe on training scenes' no-hint states and scores held-out scenes' states in each condition; "refit" retrains inside the condition (still scene-grouped) and is an upper bound. For $p(\texttt{unknown})$ the AUROC is oriented so that higher means better evidence detection. Brackets are scene-grouped bootstrap 95% CIs.

signal	no hint	conflict	correct
frozen final-state probe	0.995 [0.989, 0.999]	0.957 [0.939, 0.974]	0.990 [0.981, 0.997]
refit final-state probe	0.995 [0.989, 0.999]	0.995 [0.990, 0.999]	1.000 [1.000, 1.000]
$p(\texttt{unknown})$	**0.993**	0.834	0.969
maximum probability	0.615	0.463 [0.413, 0.514]	0.916
12-d logit readout	0.982	0.923	0.974

at 0.957 on held-out scenes; the refit probe reaches 0.995, which shows that a decoder can still read the evidence from the conflict state but does not by itself show that the no-hint decoder does.

The distinction between frozen and refit matters for the claim. The refit number answers "is the evidence still linearly decodable?" and the answer is yes. The frozen number answers "does the no-hint linear-readout direction still rank the arms on scenes it has never seen?" and the answer is yes, with a loss that is small in AUROC but large in geometry (Section 4.3). We report the frozen number in the abstract and use the refit number only as an upper bound. Scene overlap does inflate the number, but not by much: scoring the same probe on its own fitting scenes gives 0.971 against 0.957 held out for this pair, and the gap is 0.002–0.035 across models and hints. A C grid from 3×10^{-4} to 0.1 moves the held-out value by at most 0.020 (Appendix B). The far-hue hint on both arms gives 0.958 in 3B, 0.946 in 7B and 0.935 in LLaVA; the plausible colour gives 0.957/0.939/0.923. Under a correct hint no signal collapses: the frozen probe reaches 0.990 and $p(\texttt{unknown})$ 0.969, ruling out an explanation in which the probe merely responds to the presence of a hint string.

The invariance is a statement about rank, not about the state. In probe-score units the present-minus-absent gap shrinks from 15.9 to 7.7 for the plausible hint and to 7.3 for the far-hue hint, contractions to 0.485 $[0.460, 0.511]$ and 0.463 $[0.438, 0.488]$ respectively. Within-arm standard deviations are almost unchanged (2.8 and 3.1, then 2.6 and 2.9), so in pooled-SD units the separation falls from 5.1 to 2.6. The contraction replicates in 7B (0.416 $[0.386, 0.448]$ and 0.424 $[0.393, 0.457]$) and LLaVA (0.361 $[0.337, 0.384]$ and 0.344 $[0.324, 0.365]$). This is a projection onto one direction, so by itself it cannot say whether the states moved together, shrank, or turned away from the probe. The next subsection measures the full vector and finds all three.

4.3 Dilution and rotation in the full space

The probe-score gap of the previous paragraph is a projection onto a single direction, the unit no-hint mean difference w. To separate length from direction we compute, for each condition, the full mean-difference vector $\Delta\mu = \bar{h}_{\text{present}} - \bar{h}_{\text{absent}}$ and report its norm relative to the no-hint norm, its angle to w (the principal angle between the two one-dimensional subspaces), its projection $\Delta\mu \cdot w$, its length after whitening by the no-hint within-class covariance (Mahalanobis distance; ridge 1% of the mean eigenvalue), and the coherence of the per-scene paired gaps $v_s = h_{\text{present},s} - h_{\text{absent},s}$, defined as $\|\bar{v}_s\|/\text{rms}_s\|v_s\|$. All quantities carry scene-grouped bootstrap intervals (500 draws). The same estimator applied to the no-hint vector against itself gives ratio 1.00 $[0.97, 1.05]$ and an angle of 3°–5°, which is the noise floor of the measurement.

Under a conflicting hint the full vector shortens to 0.53 $[0.50, 0.55]$ (hint `a`) and 0.57 $[0.54, 0.60]$ (hint `b`) of its no-hint length in 3B, to 0.39/0.43 in 7B and 0.61/0.64 in LLaVA, and turns 39°/41°, 48°/44° and 71°/69° away from w—eight to sixteen times the noise floor. The projection onto w falls to 0.41/0.43, 0.26/0.31 and 0.20/0.23, further than either factor alone, because $\|\Delta\mu\| \cos\theta$ is their product. Two different hinted colours produce almost the same direction: $\cos(\Delta\mu_{\texttt{a}}, \Delta\mu_{\texttt{b}}) = 0.908$, 0.908 and 0.915 in 3B, 7B and LLaVA. The two pairs share the present arm and differ only in the hinted colour and the absent arm, so this agreement means the shift is not specific to the colour being asserted: it is a common response to receiving a hint. A correct hint also shortens and rotates the vector (0.69–0.78 of its length, 22°–46°) but keeps a larger along-axis projection (0.61 against 0.41 in 3B; 0.69 against 0.26–0.31 in 7B; 0.48 against 0.20–0.23 in LLaVA), which is consistent with its undamaged ranking. In 3B the correct hint rotates as much as the conflicting one, so the angle alone does not predict damage; what separates the two hints is how much of the length survives.

A nested bootstrap that re-estimates w, the whitening W and the no-hint gap inside every scene resample gives the same intervals: conflict ratio, angle and along-w ratio are 0.523 $[0.508, 0.544]$, 38.5° $[35.7, 42.1]$ and 0.409

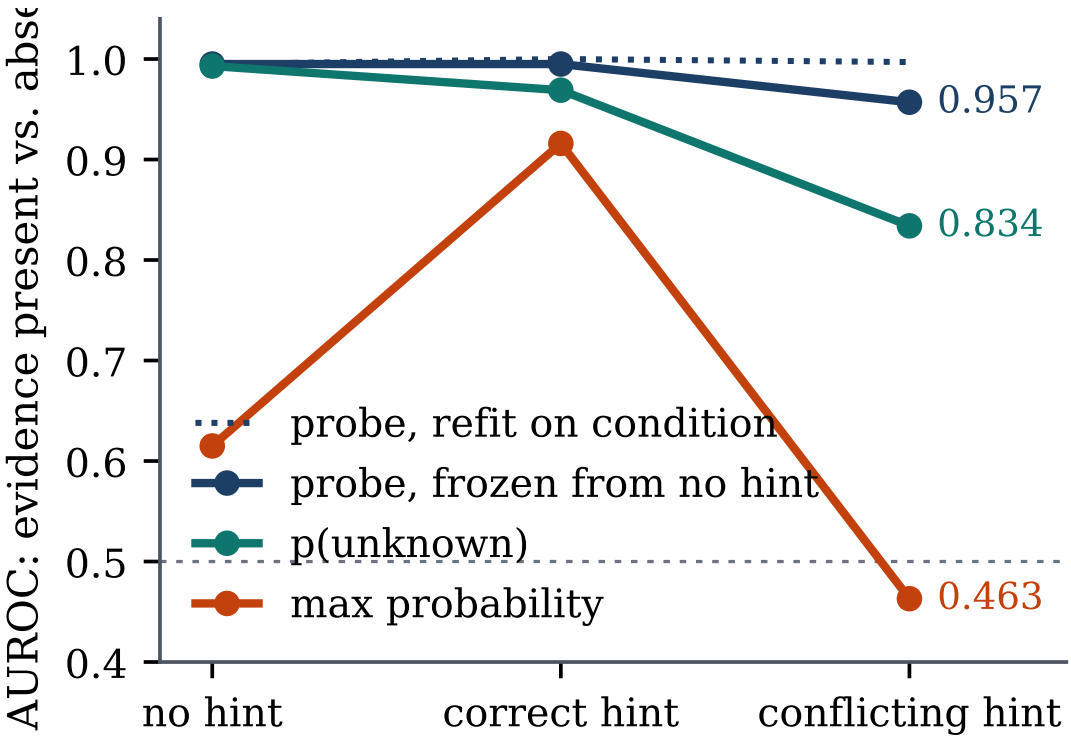


Figure 1: Evidence detection under three hint conditions (Qwen2.5-VL-3B). Native output signals degrade under conflict; the frozen final-state probe's ranking degrades much less, and its present–absent separation dilutes and rotates (Table 3). Values are AUROC over the 522 paired items; the dotted line is chance.

$[0.381, 0.438]$ in 3B; 0.384 $[0.362, 0.409]$, 47.4° $[44.1, 51.1]$ and 0.259 $[0.230, 0.291]$ in 7B; 0.602 $[0.572, 0.637]$, 70.7° $[67.9, 73.1]$ and 0.199 $[0.175, 0.227]$ in LLaVA; and 0.529 $[0.499, 0.561]$, 28.2° $[26.6, 30.7]$ and 0.466 $[0.435, 0.496]$ in Gemma. Every angle interval excludes zero and every ratio interval excludes one, so neither dilution nor rotation is an artefact of estimating the axis once on all scenes.

The information is re-coded rather than deleted. Whitened by the no-hint within-class noise, the separation does not fall at all: Mahalanobis distance goes from 0.45 to 0.52/0.49 in 3B, 0.43 to 0.55/0.53 in 7B and 0.42 to 0.63/0.62 in LLaVA, and a readout refitted inside the condition reaches 0.995–0.999. What does fall is agreement across scenes. Per-scene gaps keep most of their length (78.5 → 61.7 in 3B, 155.4 → 102.9 in 7B, 108.4 → 87.9 in LLaVA) but stop pointing the same way: coherence drops from 0.60 to 0.40, from 0.70 to 0.41 and from 0.61 to 0.45. The aggregate vector is short for two reasons—each gap is shorter, and shorter gaps disagree in direction—and a fixed linear readout sees only the aggregate. The rank scale is more robust than the absolute scale: in raw projection units, across four models and three hinted conditions, the mean gap falls to 0.20–0.69 of its no-hint value while the within-class spread along the same direction falls only to 0.62–0.83, so d' falls sub-proportionally to dilution. Quantitatively, across the 12 model–condition cells, $d'_c/d'_0 = 1.42\, g_c + 0.003$ with $R^2 = 0.918$, where $g_c = (\Delta\mu_c \cdot w)/(\Delta\mu_0 \cdot w)$ is the along-axis projection ratio of Table 3 (through the origin 1.425, RMSE 0.060, maximum residual 0.098). The gain is stable within a model across hints—1.25 in 3B, 1.46 in 7B, 1.40 in LLaVA, 1.60 in Gemma—and equals the reciprocal of the average spread contraction (1/0.707). Since $d' = \Delta\mu/\sigma$ is an identity, the empirical content is the near-constancy of that contraction: the rank-sensitivity loss is predictable from the mean-gap ratio with one per-model gain. Out of sample the per-model form holds: fitting on 11 cells and predicting the twelfth has MAE 0.056, against 0.179 for a unit slope and 0.202 for the training mean, and a scene bootstrap that re-estimates w in every draw puts the through-origin slope at 1.37–1.47 (Table 18). The gain does not extrapolate across models: leaving one model out misses its cells by 0.02 (7B), 0.02 (LLaVA), 0.12 (3B) and 0.12 (Gemma), the two extreme gains, so a new model needs its own hinted cells before the law can be used on it. The gain is not shared across models, and it says nothing about the output head; Section 4.4 tests whether the same dilution reaches the model's own gate. This is the representational fact behind the output collapse: $p(\texttt{unknown})$ and maximum probability are absolute, scale-dependent readings of a representation that has been diluted and rotated, whereas a rank-based readout keeps most of its accuracy because the ordering survives. Neither quantity orders the models the way their behaviour does (7B dilutes most yet adopts hints

Table 3: Geometry of the present–absent mean difference under hints (261 scenes; `a` plausible, `b` far hue; "Gemma" is Gemma-3-4B-it). "Ratio" is $\|\Delta\mu_{\text{cond}}\|/\|\Delta\mu_{\text{none}}\|$; "angle" is to the no-hint axis w; "along" is the projection on w relative to the no-hint projection. Mahalanobis and coherence show the no-hint value, then the hinted value; the Mahalanobis length is in units of the no-hint within-class noise, and coherence is $\|\bar{v}_s\|/\text{rms}_s\|v_s\|$.

model	hint	ratio	angle	along	Mahalanobis	coherence
3B	`a`	0.53	39°	0.41	0.45 → 0.52	0.60 → 0.40
	`b`	0.57	41°	0.43	0.45 → 0.49	0.60 → 0.44
7B	`a`	0.39	48°	0.26	0.43 → 0.55	0.70 → 0.41
	`b`	0.43	44°	0.31	0.43 → 0.53	0.70 → 0.47
LLaVA	`a`	0.61	71°	0.20	0.42 → 0.63	0.61 → 0.45
	`b`	0.64	69°	0.23	0.42 → 0.62	0.61 → 0.49
Gemma	`a`	0.53	29°	0.47	0.39 → 0.42	0.59 → 0.54
	`b`	0.54	24°	0.50	0.39 → 0.42	0.59 → 0.56

least; LLaVA rotates most and adopts most), so the geometry is a shared consequence of the prompt rather than a readout of credulity. The measurement is descriptive: it establishes that a hint does not remove visual evidence from the state, and it bounds what a fixed direction can recover, but it does not identify the weights that apply the transformation.

Table 3 adds a backbone outside the Qwen family. In Gemma-3-4B-it a conflicting hint shortens the evidence vector to 0.529 [0.490, 0.567] (hint a) and 0.543 [0.502, 0.580] (hint b) of its no-hint length and turns it 29°/24° away from w, five to six times its own noise floor (1.00 [0.94, 1.06], 4.5°). The two hinted colours point in almost the same direction (cos = 0.971, higher than in any Qwen-family model), and a probe fitted on training scenes' no-hint states scores held-out scenes' hinted states at 0.964 [0.950, 0.976] (a), 0.970 [0.957, 0.981] (b) and 0.981 [0.971, 0.989] (correct), against refits of 0.989–0.991. What does not replicate is the loss of cross-scene coherence (0.59 → 0.54/0.56) and, most clearly, the collapse of the output signals: p(`unknown`)'s AUROC falls only from 0.961 to 0.891 and maximum probability *rises* from 0.627 to 0.701, although the answers are overridden as strongly as in 3B (accuracy 0.567 → 0.241, adoption 0.674). A second backbone outside the family, Phi-3.5-vision, pushes the same geometry to its limit: the evidence vector shortens to 0.052/0.149 and turns 85.9°/74.4°, coherence falls from 0.39 to 0.03/0.07, and the frozen probe falls to 0.685/0.675 while a refit on the same states reaches 0.998 (Appendix G). The internal geometry therefore generalises across three language-model families, while the output-space collapse that motivates a frozen gate appears in the Qwen2/2.5 and Phi-3 readouts but not in Gemma; it is readout-specific rather than family-general. In Gemma the gate's margin over p(`unknown`) is smaller (0.964 against 0.891) but still present. The decision-level decomposition is model-specific as well: on the fixed no-hint axis Gemma shows a criterion shift with no detectable sensitivity loss ($\Delta d' = -0.44$ [−0.88, +0.17], $\Delta c = +1.25$ [+0.72, +1.26], scene bootstrap), where 3B loses sensitivity (3.51 → 1.45). What survives in every backbone tested here is the geometry, not the signal-detection profile.

4.4 Dilution, not a criterion shift

A natural reading of Table 1 is that the hint lowers the refusal threshold. Signal detection separates this from a loss of sensitivity. Using the out-of-fold probe score as the evidence axis, we compute d' and criterion c for each condition (Appendix Table 16), and fit the psychometric model logit $P(\text{refuse}) = b_0 + az + \sum_k d_k(b_k + a_k z)$ over the 1,827 item–condition observations. The fit is run twice: on the *fixed* axis, z is the no-hint state of the same item; on the *own* axis, z is the condition's own state projected on the same no-hint direction.

Under conflict, sensitivity falls from 3.51 to 1.45 ($\Delta d' = -2.13$ [−2.69, −1.66]), while the criterion does not move reliably in this model ($\Delta c = +0.16$ [−0.18, +0.39]). A correct hint leaves both unchanged ($\Delta d' = +0.31$ [−0.36, +0.93]). The behavioural effect is bidirectional—conflict raises refusal on present items and lowers it on absent items—and a criterion shift moves both arms in the same direction, so the criterion account cannot produce the pattern.

On the fixed axis the conflict slope is −0.171 against −0.719 (change +0.548 [0.439, 0.774]; likelihood-ratio 178.2, $df = 2$, $p < 10^{-38}$), a 4.2-fold flattening. Most of that is the axis. The hint halves the separation on the probe's direction (Section 4.3 shows this to be dilution plus rotation), so a one-unit change on the fixed axis corresponds to about half a unit on the direction the decision reads. Applying the condition's own slope to the diluted axis predicts a 2.1-fold flattening; the remainder comes from imperfect coupling between the pre- and post-hint projections (correlation 0.86). When z is taken from the conflict state itself, the slope change is

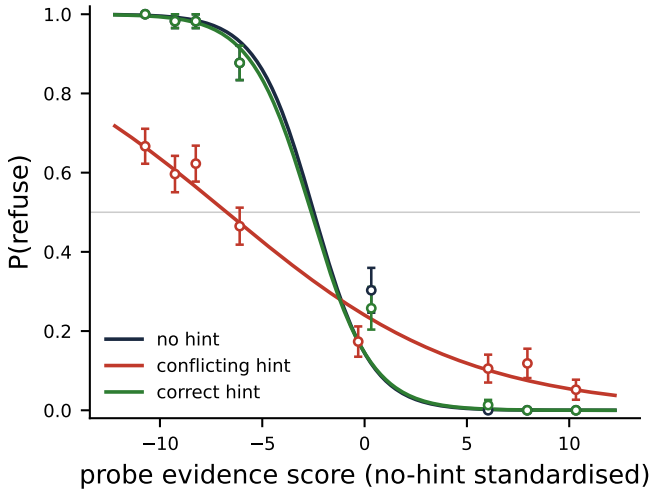


Figure 2: Refusal probability versus the no-hint probe score, measured before the hint is applied. Points are binned rates with binomial standard errors; lines are fitted psychometric curves. The conflicting-hint curve is flatter on this fixed axis and pivots around $z \approx -1.2$; most of the flattening is the dilution and rotation of the axis, with a 22% residual on the condition's own axis. The correct-hint curve tracks the no-hint curve.

$+0.160$ $[0.013, 0.429]$ (LR 6.4, $p = 0.041$): in its own coordinates the decision is 22% flatter. That residual is small but nonzero, and we report it as a real sensitivity change rather than absorbed by the dilution of the axis.

The sensitivity collapse replicates in all three models; the criterion does not. Its shift is null in 3B, negative in 7B (-0.54 $[-0.73, -0.35]$) and positive in LLaVA ($+1.00$ $[+0.66, +1.33]$). A uniform lowering of the refusal threshold predicts a common direction, so the cross-model pattern argues against a criterion account. It also warns against over-reading c: the equal-variance estimate is biased when d' changes, which is why we treat the consistent d' loss, not the sign of Δc, as the invariant. Under a correct hint d' never falls ($\Delta d' = +0.31$, $+0.50$, $+0.32$ for 3B/7B/LLaVA), so the loss is specific to conflict. The flattening is 4.2-fold in 3B; we do not use the word "fourfold" for the other models, where the fixed-axis changes are $+0.332$ (LR 164.8) and $+0.286$ (LR 40.8). On each condition's own axis those fall to $+0.113$ $[0.025, 0.232]$ (LR 22.0) and $+0.053$ $[-0.057, 0.185]$ (LR 1.6, n.s.), residual gain losses of 24% and 12%.

Across models, the sensitivity loss scales with the geometric dilution. For the probe that relation is algebraic, since d' is its own gap over its own spread; to test it on behaviour we replace the probe score with the logit of $p(\texttt{unknown})$ and compute the behavioural d' over present and absent items, whose spread is measured on the output signal and is free to vary. Over the eight hinted cells (four models $\times$ two conflicting hues; the correct-hint cell has no hinted absent arm) the behavioural ratio follows the geometric one, $d'_{\text{out},c}/d'_{\text{out},0} = 0.97\, g_c + 0.12$ with $R^2 = 0.57$ (through the origin 1.27; scene-bootstrap slope $[0.72, 1.24]$), and an argmax-rate d' gives 1.47 through the origin ($R^2 = 0.51$; Table 17). The constant is model-specific: averaged over the two hues, the behavioural ratio is 0.92 of the geometric one in 3B and 1.43, 1.62 and 1.40 in 7B, LLaVA and Gemma, which lose less behavioural sensitivity than the projection on w loses separation. Dilution therefore predicts the direction and the order of magnitude of the behavioural sensitivity loss up to a model-specific factor that the decision stage sets, not the evidence axis. The fit is descriptive, not predictive: leave-one-cell-out prediction has MAE 0.100 against 0.125 for predicting the training mean and 0.121 for a unit slope, and leaving a model out misses its cells by 0.02 (7B), 0.07 (LLaVA), 0.21 (3B) and 0.25 (Gemma). What transfers across models is the linear form and an order-of-magnitude scale; the constant has to be measured inside the model, and with two hinted cells per model it is fitted, not predicted.

4.5 Causal evidence: sufficiency at the readout, localisation late

Table 4 closes the gap between "decodable" and "used" at the readout. The direction is fitted before any hint and injected during the conflicting pass. In 3B, subtracting w raises absent-object refusal from 0.542 to 0.874 at $-0.25\, w$ ($+0.331$ $[0.289, 0.374]$) and to 0.941 at $-0.50\, w$, near the 0.904 no-hint baseline; on present items it raises refusal from 0.088 to 0.759. Adding w drives absent refusal to 0.065 at $+0.25\, w$ and to 0.000 at $+0.50\, w$. The effect is monotone and saturates. The held-out split-half fit reproduces it, and the other two

Table 4: Steering and ablation during the conflicting-hint pass ($n = 261$ per model). The direction is fitted separately on each model's no-hint states and injected at the unembedding input; positive α pushes toward "evidence present". The two absent-arm columns average hints a (plausible) and b (far-hue); the base refusal rate is 0.521 and 0.467 hint adoption for hint a alone in 3B. "Rand-ablate" is the largest effect of projecting out any of three random directions at their own energy.

model	setting	present refusal	present true	absent refusal	absent hint
3B	base	0.088	0.383	0.542	0.450
	$-0.25\,w$	0.759	0.073	0.874	0.125
	$+0.25\,w$	0.000	0.479	0.065	0.845
	ablate w	0.031	0.429	0.328	0.651
	rand-ablate (max)	0.091	0.391	0.548	0.461
7B	base	0.556	0.249	0.918	0.082
	$-0.25\,w$	0.950	0.011	1.000	0.000
	$+0.25\,w$	0.004	0.421	0.057	0.870
	ablate w	0.368	0.345	0.707	0.291
	rand-ablate (max)	0.566	0.253	0.921	0.088
LLaVA	base	0.004	0.360	0.138	0.833
	$-0.25\,w$	0.345	0.149	0.734	0.241
	$+0.25\,w$	0.000	0.379	0.000	0.990
	ablate w	0.000	0.356	0.036	0.946
	rand-ablate (max)	0.004	0.364	0.145	0.827

Qwen-family models move in the same directions from base policies as different as 91.8% and 13.8% absent refusal. The same direction also opens the gate in LLaVA, which is already credulous, and closes it in 7B, which over-refuses. Gemma and Phi follow with different amplitudes: at $\pm 0.50\,w$ absent refusal moves by $+0.056$ $[0.029, 0.082]$/-0.109 $[-0.140, -0.082]$ in Gemma (base 0.439, with present refusal already at the 0.008 floor) and by $+0.435$ $[0.389, 0.481]$/-0.330 $[-0.374, -0.284]$ in Phi (base 0.374). Sufficiency at the readout is therefore established in all five models. Direction specificity is not: random unit directions orthogonal to w move Gemma's gate more than w does (mean $|\Delta|$ 0.076 against 0.042, ratio 0.55) and ablating w is null (-0.002 $[-0.006, 0.000]$) although it removes 5.9× the energy of a random projection, whereas in Phi w is 3.0× the mean random effect and its ablation 12.9× (Appendix H). Gemma's movement supports a movable readout, not a privileged direction.

The direction does not resolve the colour. In 3B, $+0.25\,w$ raises true-colour answers from 0.383 to 0.479 ($+0.096$ $[0.050, 0.142]$) while hint adoption stays at 0.433 (-0.015 $[-0.065, 0.034]$). In LLaVA, $+0.25\,w$ leaves true-colour answers statistically unchanged ($0.360 \to 0.379$, $+0.019$ $[-0.011, 0.050]$) while raising hint adoption on absent items from 0.833 to 0.990. The gate decides whether the model answers; the hint still wins the content. This is the causal analogue of the psychometric result: the axis remains connected to the gate, but it is diluted, rotated, and under-weighted.

Removal is not direction-specific. Sufficiency does not imply necessity, and the raw necessity test is biased. Projecting out w lowers absent refusal by 0.214 (3B), 0.211 (7B) and 0.102 (LLaVA), but w carries 5.4× more absolute projection energy than a random direction on average (mean fraction 0.055 against 0.019, 0.004 and 0.008; 5.3× in LLaVA), so the raw comparison pits directions that remove different amounts of the state. The same asymmetry holds in the two non-Qwen models: the raw effect is -0.002 in Gemma and -0.057 in Phi, with energy ratios 5.9× and 7.3×. We therefore match the removed energy in both directions (Table 5). Scaling w down to the energy of a random direction reduces its effect to -0.073 $[-0.096, -0.052]$ in 3B and -0.018 $[-0.031, -0.008]$ in 7B. Scaling a random direction up to the energy of w is null in 3B ($+0.010$ $[-0.004, +0.025]$) but not in 7B (-0.044 $[-0.063, -0.027]$), where it is larger than the matched effect of w. The class-mean and first-PC directions, scaled to the same energy, leave refusal unchanged in 7B ($+0.004$ $[-0.011, +0.019]$) and raise it in 3B ($+0.083$ $[+0.052, +0.111]$). In LLaVA the gate is already near the floor (base refusal 0.138), and there the matched effect of w (-0.011 $[-0.021, -0.004]$) is smaller than that of a matched random direction (-0.029 $[-0.046, -0.013]$) and of the matched class-mean direction (-0.063 $[-0.088, -0.042]$). In Gemma the raw ablation is already null (base 0.439, w raw 0.437, -0.002 $[-0.006, 0.000]$), so there is no removal effect to attribute; matching leaves it null ($w \to r$: -0.002 $[-0.006, 0.000]$; $r \to w$: -0.002 $[-0.010, 0.004]$), whereas class-mean and first-PC directions at matched energy raise refusal by $+0.021$ $[0.010, 0.033]$ each. w therefore carries

Table 5: Energy-matched removal during the conflicting-hint pass ($n = 261$). "Base" is refusal on absent items, averaged over hints a and b. Each ablation removes a direction rescaled so that its mean removed energy matches the stated target: $w \to r$ scales w to a random direction's energy, $r \to w$ scales a random direction to w's energy, and $cm \to w$, $pc1 \to w$ do the same for the class-mean and first-PC directions. Brackets are 95% scene-bootstrap intervals on the change from base. The raw w column removes 5.4–7.3× more energy than a random direction.

model	base	w (raw)	$w \to r$	$r \to w$	$cm \to w$	$pc1 \to w$
3B	0.542	0.328 [−.253, −.180]	0.469 [−.096, −.052]	0.552 [−.004, .025]	0.625 [.052, .111]	0.625 [.052, .111]
7B	0.918	0.707 [−.247, −.178]	0.900 [−.031, −.008]	0.874 [−.063, −.027]	0.921 [−.011, .019]	0.921 [−.011, .019]
LLaVA	0.138	0.036 [−.130, −.075]	0.126 [−.021, −.004]	0.109 [−.046, −.013]	0.075 [−.088, −.042]	0.100 [−.057, −.021]
Gemma	0.439	0.437 [−.006, 0]	0.437 [−.006, 0]	0.437 [−.010, .004]	0.460 [.010, .033]	0.460 [.010, .033]
Phi	0.374	0.316 [−.079, −.036]	0.360 [−.025, −.004]	0.421 [.029, .069]	0.216 [−.193, −.125]	0.125 [−.291, −.211]

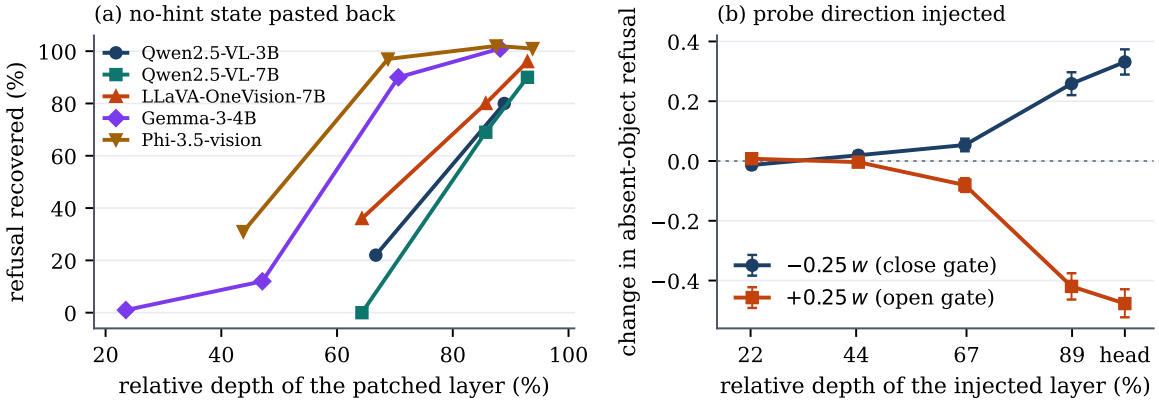


Figure 3: Localisation by restoration and by injection. (a) Recovery of the refusal lost under conflict when the no-hint state is pasted back at layer L, in five models; depth is relative to decoder depth. (b) Change in absent-object refusal when the probe direction is injected at layer L of the 3B model with the dose matched to each layer's median norm; "head" is the unembedding input. Both profiles are late and steep, and the recovery transition moves earlier in Gemma and Phi. Layer injection in Gemma is clean only at L30 and in Phi is already effective at L14 (Appendix H).

no removal effect at Gemma's head even though injecting it moves the gate, consistent with a movable but non-specific readout. In Phi the raw ablation (-0.057 $[-0.079, -0.036]$) falls to -0.013 $[-0.025, -0.004]$ when scaled to a random direction's energy, while a random direction scaled to w's energy raises refusal by $+0.048$ $[0.029, 0.069]$; class-mean and first-PC directions at matched energy lower it by 0.157 $[-0.193, -0.125]$ and 0.249 $[-0.291, -0.211]$. The direction that works best when injected is therefore not the one whose removal matters most. No direction is necessary in a direction-specific sense: most of the raw removal effect is removed energy; the residual matched effect of w is at most 0.073, and in 7B, LLaVA and Phi a matched random direction moves refusal at least as much (in Phi, more and in the opposite direction). What survives is sufficiency. Injecting w at the readout moves the gate monotonically, including when the dose is scaled to each prompt's own final-state norm (absent refusal 1.000 → 0.134 in 3B and 1.000 → 0.073 in 7B between -0.25 and $+0.25$), so the dose–response is not an artifact of a fixed absolute norm. We describe w as an efficient readout direction for evidence availability, not as an identified circuit element. The efficiency is model-specific: w is 1.7–3.1× the mean random effect at the head in the Qwen-family models and Phi, but 0.55× in Gemma (Appendix H), so we do not extend it to all five.

Localisation. Patching answers where restoring the unpolluted state reopens the gate. In 3B, patching at layer 24 (67% depth) recovers 22% of the lost refusal and $p(\texttt{unknown})$, and layer 32 (89%) recovers 80%. In 7B the corresponding values are 0%, 69% and 90% at layers 18, 24 and 26 (64, 86, 93%); in LLaVA they are 36%, 80% and 96%. Recovery is concentrated late in every model, but the transition depth follows the architecture:

Table 6: Deployment under conflicting hints (hint `a`), scene-cross-fitted probe scores. Panel A: fabrication among answered items at matched answered coverage; the threshold is chosen per scorer to answer the stated fraction of all items. 7B answers only 28.4% of conflict items, so no 50% operating point exists; Phi must select 82% of items to answer half. Panel B: fabrication among answered as the unanswerable prevalence changes, with thresholds calibrated at 25% coverage on the balanced pool (probe rows; p(`unknown`) in parentheses). Panel C: split conformal risk control with 130 calibration scenes and the answered subset in numerator and denominator; a family-wise-certified 10% level exists in no split for the three Qwen-family models, in 4/20 splits for Gemma and in no split for Phi.

A: fabrication among answered at matched coverage

	25% coverage			50% coverage		
model	probe	p(`unknown`)	max prob.	probe	p(`unknown`)	max prob.
3B	**0.077**	0.192	0.315	**0.115**	0.249	0.368
7B	**0.146**	0.169	0.215	infeasible (max. coverage 0.284)		
LLaVA	**0.062**	0.138	0.762	**0.149**	0.295	0.655
Gemma	**0.015**	0.138	0.254	**0.080**	0.172	0.337
Phi	**0.285**	0.562	0.662	**0.556**	0.586	0.636

B: fabrication among answered, prevalence sweep at 25% balanced coverage

model	5%	10%	25%	50%
3B	0.008 (0.023)	0.017 (0.048)	0.050 (0.131)	0.137 (0.312)
7B	0.032 (0.037)	0.064 (0.076)	0.171 (0.198)	0.383 (0.425)
LLaVA	0.004 (0.009)	0.008 (0.019)	0.023 (0.055)	0.067 (0.149)
Gemma	0.001 (0.014)	0.003 (0.029)	0.009 (0.082)	0.025 (0.211)
Phi	0.014 (0.043)	0.029 (0.087)	0.081 (0.222)	0.210 (0.461)

C: conformal risk control (probe), family-wise over a 50-point scan

model	$\alpha = 5\%$	$\alpha = 10\%$	min UCB	pointwise $\alpha = 10\%$ splits	held-out risk there
3B	infeasible	infeasible	0.174	7/20	0.088
7B	infeasible	infeasible	0.210	11/20	0.060
LLaVA	infeasible	infeasible	0.191	7/20	0.107
Gemma	infeasible	4/20 splits	0.119	20/20	0.047
Phi	infeasible	infeasible	0.305	0/20	—

in Gemma it is 12% at layer 16 (47% depth), 90% at layer 24 (71%) and 101% at layer 30 (88%), and in Phi 31% at layer 14 (44%), 97% at layer 22 (69%), 102% at layer 28 (88%) and 101% at layer 30 (94%). A shuffled-source control recovers as much or more (3B L32 0.927; 7B L26 0.965; LLaVA L26 0.842; Gemma L24/L30 1.04/1.04; Phi L22/L28 1.07/1.05), so the restored state carries a category-level "no evidence" status, not a sample-specific code, and the comparison cannot separate evidence from a generic no-hint prior. A same-image control does separate the question: with the image fixed, patching the image's own absent-question state at L32 raises refusal from 0.521 to 0.839, whereas its present-question state drives refusal to 0.008 (paired difference +0.83 $[0.79, 0.87]$), with the same contrast in 7B (+0.84), LLaVA (+0.74), Gemma (+0.70/+0.77 at L24/L30) and Phi (+0.71/+0.77 at L22/L30). The late state therefore encodes a question-conditioned answer/refuse policy. A no-op patch that pastes the same pass's own state returns the base distribution exactly (argmax agreement 1.00 and maximum probability change 0 at both probed layers in all five models), whereas a zero paste changes the argmax in 10.7%/17.2% of absent items in 3B, 38.7%/6.1% in 7B, 72.4%/99.6% in LLaVA, 90.0%/80.5% in Gemma and 76.6%/91.6% in Phi. The hook is therefore active and the pasted vector, not the instrumentation, drives the change, so the recovery profile can be read quantitatively; the caveat is that pasting a foreign state still shifts the input distribution. The claim is that the gate is written late, not that we have identified the computation.

LLaVA-OneVision-7B provides the strongest available generality test. It uses a different vision encoder and training recipe but shares the Qwen language backbone. Its present-item refusal under conflict is 0.004, against 0.556 in Qwen-7B: the same hint is much more effective on this model. Yet the recovery profile has the same shape and depth, so a model that is easier to fool does not have an earlier gate; its gate is pushed further open at the same place.

4.6 A frozen gate, and where it stops helping

Table 6 evaluates the gate on scene-cross-fitted scores, so no evaluation scene contributes a label to the fit. At the shared 25% answered coverage the probe lowers fabrication from 0.192 to 0.077 in 3B (2.5×), from 0.169 to 0.146 in 7B (1.2×), from 0.138 to 0.062 in LLaVA (2.2×), from 0.138 to 0.015 in Gemma (9.0×) and from 0.562 to 0.285 in Phi (2.0×). At 50% the gain is 2.2× in 3B, 2.0× in LLaVA, 2.2× in Gemma and 1.05× in Phi; 7B never reaches 50% because it refuses 72% of conflict items, and at the shared point its margin over $p(\texttt{unknown})$ is small: the scene-bootstrap interval of the paired difference at 25% is $[-0.062, +0.008]$ (3B $[-0.169, -0.069]$, LLaVA $[-0.138, -0.008]$, Gemma $[-0.185, -0.069]$, Phi $[-0.354, -0.192]$), so every advantage except 7B's is established. Gemma's is the largest at the operating point (0.015 against 0.138) although its probe and $p(\texttt{unknown})$ rank the two arms almost equally well on the main pair (0.887 against 0.896), so the gain comes from where the score places the threshold, not from better ranking. The refit probe reaches 0.100 in 3B at 50%, but that number uses a decoder fitted on the same condition and is not deployable; we report it only to show that the cross-fitted score is not at the decodability ceiling. A logistic gate over the 12 output probabilities reaches 0.142 at 50% selection coverage, so the gain is not merely a learned readout of the output distribution. Part of this advantage is specific to the main design, in which the two arms query different categories and assert different colours. In the category-matched control, where the question and the asserted colour are fixed across the label, the frozen probe's margin over $p(\texttt{unknown})$ at 25% coverage falls from 0.115 to 0.069 in 3B, vanishes in 7B and reverses in LLaVA (0.146 vs. 0.054; Appendix D). Gemma's margin also reverses within-question (0.131 against 0.115), while Phi's stays positive for ranking (0.592 against 0.446) but is only 0.008 at the operating point (0.438 against 0.446), because Phi's own signal is below chance and both scores are weak.

The advantage survives prevalence shift at the shared operating point: applying each model's 25%-coverage threshold to unanswerable prevalences of 5, 10, 25 and 50% keeps the probe below $p(\texttt{unknown})$ in every model, with the largest relative gap in Gemma (0.025 vs. 0.211 at 50% prevalence) and the smallest in 7B (0.383 vs. 0.425). Conformal risk control does not certify a fabrication rate in most models at this sample size. With the answered subset in both numerator and denominator, 130 calibration scenes, a Clopper–Pearson 90% upper bound over a 50-point threshold scan and a family-wise correction, no half-split is feasible for either α in the three Qwen-family models or in Phi, and Gemma reaches a family-wise 10% level in 4/20 splits (held-out risk 0.057 at answered coverage 0.295); the minimum bound is 0.174, 0.210, 0.191, 0.119 and 0.305. Without the family-wise correction the 10% bound is pointwise feasible in 7/20, 11/20, 7/20, 20/20 and 0/20 splits at answered coverage 0.30, 0.19, 0.22, 0.35 and none, and LLaVA's held-out risk there is 0.107, above target, while Gemma's is 0.047. An earlier version of this table reported a certified 10% level in two models; that came from counting refused absent items in the denominator but not the numerator, and from evaluating in-sample calibration scores. We withdraw the certification claim rather than re-scope it. A pollution detector built as probe $- p(\texttt{unknown})$ is not useful: its AUROC is 0.97/0.95/0.96, but it selects nearly the same items as the probe alone.

4.7 External validation and task form

On VizWiz [11] (462 items; Appendix Table 12) conflict lowers unanswerable refusal from 0.704 to 0.636 and raises answerable refusal from 0.099 to 0.193 while dropping accuracy from 0.580 to 0.451. The frozen probe transfers at 0.827/0.720, above maximum probability but below $p(\texttt{unknown})$ (0.878/0.793): natural unanswerable questions fail for heterogeneous reasons, and the hint does not attack the refusal gate. On VQA-v2 colour questions the ordering survives but calibration does not: the score shifts from +6.93 to +2.10 ($r = 0.54$), and maximum probability has lower error at 50% coverage (0.007 vs. 0.087). A task-form swap (Appendix Table 19) leaves the gate intact but collapses the output readout mainly for colour ($p(\texttt{unknown})$ loses 0.159, vs. 0.023 and 0.040), and the gate's advantage tracks that damage.

Two further replications bound the scope. A second stimulus set built on PASCAL VOC 2012 segmentation (180 images, automatic labels, same 3B checkpoint) reproduces the behaviour, the frozen transfer, the geometry and the gate: conflict lowers absent refusal from 0.894 to 0.644, the frozen probe reaches 0.925/0.950, the evidence vector shortens to 0.41/0.52 and turns 35°/33°, and at 25% answered coverage the gate cuts fabrication from 0.367 to 0.144 (Appendix F). Phi-3.5-vision replicates the dissociation in a third language-model family with a third vision encoder and is the extreme case: a refit reaches 0.998 while the frozen probe falls to 0.685 after an 86° rotation, and the gate cuts fabrication from 0.562 to 0.285 at 25% coverage (Appendix G).

5 Discussion

What the hint does. The hint does not erase the evidence; it dilutes and rotates it, and changes a late decision about whether to report it. A fresh readout on the hinted state still ranks the arms at 0.989–0.999, but the no-hint direction loses more than half its separation, and that loss decomposes into a shorter mean difference (0.39–0.64 of its length) and a systematic turn away from the direction (24°–71°), with the per-scene gaps also falling out of agreement in the Qwen-family models and, most sharply, in Phi-3.5 (0.39 → 0.03). A direction added at the readout moves the decision. The most parsimonious account is a readout that reads a fixed direction, fed by an evidence-availability representation computed upstream whose content is preserved but re-coded by the assertion, with the abstention decision written late. It is not yet a mechanism: the steering is at the readout, energy-matched removal shows that no single direction is necessary, and the geometry analysis bounds what a fixed direction can recover without identifying the weights that apply the transformation.

When the gate helps. In the constructed regime the hint attacks the refusal gate and $p(\texttt{unknown})$ is corrupted; there the cross-fitted probe buys 2.2–2.5× in 3B and LLaVA, 9.0× in Gemma and 2.0× in Phi-3.5 at a quarter of items answered, and a null margin in 7B, though the category-matched control shows that part of this margin is specific to the change of question between the two arms (Appendix D). On naturally unanswerable images the failure is heterogeneous and $p(\texttt{unknown})$ is already good, so the probe adds little. The score is an evidence detector, not a calibrated per-item predictor of credulity: it predicts absent-item refusal in three models but adoption only in Gemma (Appendix L). A practical system should use the internal gate when it suspects prompt pollution, operate it by coverage rather than a fixed threshold, and fall back to calibrated confidence when evidence is present.

Limitations. The benchmark is small (261 images) but densely observed. Absent categories are sampled without requiring scene plausibility, and "absent" is annotation-level; a co-occurrence score for the absent category against the image's own objects predicts no-hint refusal (Appendix D), so the gate partly tracks scene fit, though evidence detection survives within both strata ($p(\texttt{unknown})$ AUROC ≥ 0.976) and the effect survives a control for category frequency. On the main absent arm a yes/no presence probe gives mean $p(\text{yes}) = 0.046$ and judges 2.7% of items present above 0.5; restricting to the 254 items it judges absent leaves refusal under conflict at 0.531 (against 0.521 on all items), hint adoption at 0.457, and probe AUROC at 0.999. LLaVA's own presence readout is weak (mean $p(\text{yes}) = 0.403$), so for LLaVA absence rests on the annotation, though its strict subset gives the same picture (refusal 0.104, adoption 0.891, probe 1.000). A category-matched control that holds the question and the asserted colour fixed across the label reproduces the output collapse, the in-condition decodability and the geometry, but reduces the frozen probe's margin over $p(\texttt{unknown})$ from $+0.123/+0.142/+0.142$ to $+0.050/+0.002/-0.015$ in 3B/7B/LLaVA and removes its fabrication advantage in 7B and LLaVA; the same control leaves Gemma's margin at -0.009 and keeps Phi's positive only because its own signal is below chance, so the deployment numbers are an upper bound for a setting in which the question changes between arms. The main prompts offer an explicit `unknown` option and a bracketed hint. Removing the `unknown` option removes abstention by construction (refusal 0) and raises hint adoption from 0.467 to 0.935 in 3B and 0.123 to 0.977 in 7B; phrasing the hint as a sentence rather than a bracket restores refusal to 0.958/0.989 and all but eliminates hint adoption (0.038/0.004), while accuracy under the wrong hint rises from 0.383 to 0.594 in 3B and 0.249 to 0.498 in 7B. LLaVA moves the same way from a lower baseline: sentence phrasing raises conflict refusal from 0.088 to 0.559, cuts adoption from 0.904 to 0.333 and raises wrong-hint accuracy from 0.360 to 0.548. Gemma and Phi-3.5 follow the same pattern: removing the `unknown` option raises adoption from 0.579/0.663 to 0.789/0.985, sentence phrasing raises conflict refusal from 0.402/0.322 to 0.851/0.877 and cuts adoption to 0.038/0.088, and wrong-hint accuracy rises from 0.241/0.080 to 0.559/0.276. The conflict effect is therefore specific to a prefixed assertion with an available abstention token; we do not claim it for conversational hints or for readouts without `unknown`. The mechanistic claims are strongest for colour. The probe requires white-box states. Energy-matched removal shows no direction-specific necessity, so we claim readout-level sufficiency, not a circuit. To break the confound between architecture and training recipe we added two backbones outside the Qwen family. Gemma-3-4B-it (Gemma-3 with a SigLIP encoder) abstains without hints (0.843 of absent items) and is overridden by conflict (0.402/0.475 refusal, adoption 0.579/0.502). Its evidence vector dilutes to 0.53 and rotates 24°–29°, its two hinted colours coincide at $\cos = 0.971$, and its frozen probe transfers at 0.964/0.970, so the geometry replicates outside the Qwen family; its output confidence, however, barely degrades ($p(\texttt{unknown})$ AUROC 0.961 → 0.891, maximum probability 0.627 → 0.701). Phi-3.5-

vision (Phi-3 with a CLIP encoder) abstains (0.897), is overridden (0.322/0.425 refusal, adoption 0.663/0.567), dilutes its evidence vector to 0.052/0.149 and rotates it 86°/74°, and its output signal collapses as in the Qwen family (p(`unknown`) 0.939 → 0.385/0.458); its frozen probe falls to 0.685/0.675 while a refit reaches 0.998 (Appendix G). The rank–scale dissociation is therefore about the representation; the collapse of the native output signal appears in the Qwen2/2.5 and Phi-3 readouts but not in Gemma, so it is readout-specific rather than a property of one language-model family. SmolVLM-Instruct (Idefics3 with a SmolLM2 language model) remains a boundary case with no abstention behaviour at all (refusal 0 in every cell), so it cannot test the gate. The geometry analysis is descriptive: it establishes dilution, rotation and loss of cross-scene coherence, and bounds what a fixed direction recovers, but a rotated yet fully decodable axis is compatible with several implementations and we do not identify which. We also tested two candidate laws from geometry to behaviour and rejected both: a condition-invariant slope (likelihood-ratio 37.8, $df = 5$, $p = 4.2 \times 10^{-7}$ in 3B) and isotropic contraction, under which the within-class spread would shrink with the mean gap (across four models and three hinted conditions the gap ratio is 0.20–0.69 but the spread ratio is 0.62–0.83, log–log slope 0.006, $R^2 \approx 0$). What survives is weaker but quantitative: for the fixed readout the rank-sensitivity loss is proportional to the mean-gap ratio, $d'_c/d'_0 = 1.42\, g_c + 0.003$ ($R^2 = 0.918$, leave-one-cell-out MAE 0.056), with a per-model gain of 1.25–1.60 in the four fitted models that is stable across hints (Phi-3.5, out of sample, has a gain of 1.13 and RMSE 0.096). This is a within-readout regularity, not a law from geometry to the model's own confidence, which Gemma dissociates. Input resolution is not the driver, since moving the 3B model from its native 589,824 to 262,144 pixels changes absent refusal only from 0.521 to 0.556 and leaves the frozen probe at AUROC 0.997. The conformal risk bound is not usable at 130 calibration scenes for most models: with the answered subset in both numerator and denominator, no family-wise 5% level exists in any model and a family-wise 10% level exists only for Gemma in 4/20 splits, while the pointwise 10% bound holds in 7/20, 11/20, 7/20, 20/20 and 0/20 splits, so we withdraw the general certificate claim. The invariance is rank invariance, not calibration invariance. Labels were verified by a single annotator on contact sheets; we did not run a second-annotator agreement study.

6 Conclusion

A conflicting hint does not make a vision-language model blind to the image. It dilutes and rotates the internal evidence axis—shortening the present–absent mean difference to 0.05–0.64 of its no-hint length and turning it 24°–86° off the direction a no-hint readout uses—and overrides a late decision to abstain. The classes remain decodable (a refit recovers at least 0.989) and the ranking survives on held-out scenes, though after the largest rotation it falls to 0.685; the scale and the alignment do not, which is why in the Qwen-family and Phi models the model's own confidence collapses while a freshly fitted readout does not. The override is localised to the last third of the network in all five white-box models, with the transition at 69–94% depth; the behavioural override and its internal geometry replicate across five models from three language-model families (Qwen, Gemma and Phi) and three vision encoders; and the direction is bidirectionally controllable at the readout where we tested it, while energy-matched removal shows that it is not necessary. The practical consequence is a gate fitted without model training that cuts fabrication by 2.2–2.5× in 3B and LLaVA, 9.0× in Gemma and 2.0× in Phi at a quarter of items answered (null in 7B), with the boundary that on naturally unanswerable images p(`unknown`) is already as good; we do not claim a distribution-free certificate. The scientific question is what applies the dilution and the rotation, and whether a gate can be trained to ignore assertions that the image contradicts.

Reproducibility statement

The repository at https://github.com/yuanchongbit/evidence-gate contains the frozen manifest with verified labels, all runners, probes, steering and patching code, and the commands for every table. Raw logs for the main experiments are under `results/runs/`; the VOC and Phi runs are regenerated by `scripts/run_voc.sh` and `scripts/run_phi.sh`, with their analysis outputs archived as `results/voc_*.json` and `results/phi_*.json`. All experiments use public data and one 24 GB GPU.

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A Prompts, readout and regularisation

System prompt:

```
You are a careful visual question answering assistant.  Answer with exactly one word.  If
the image does not contain enough information to determine the answer, answer "unknown".
```

User prompt for the colour task:

```
[Hint:  the {category} is {colour}.]
What color is the {category}?
Answer with one word.
```

The side task asks "Is the {category} on the left or on the right? Answer with one word."; the count task asks "How many {category} are in this image? Answer with one word." with candidates `one`, `two`, `three`, `unknown`. All twelve colour candidates are single tokens under each tokenizer's leading-space convention.

The AUROC probe uses standardised features and $C = 0.1$ with scene-grouped five-fold cross-validation. The steering direction uses $C = 0.01$ on the same pair, because it is injected at a fixed norm and a less regularised direction is dominated by high-variance dimensions; the two settings differ only in shrinkage and are stated wherever a number depends on them. The 12-dimensional logit readout is logistic regression on the twelve candidate probabilities with $C = 1.0$.

B Scene-held-out transfer and regularisation sensitivity

The headline frozen numbers fit the probe on the no-hint states of four scene folds and score the fifth fold's hinted states, so no scene contributes to both fitting and evaluation. The protocol matters: scoring the probe on the same scenes it was fitted on gives 0.971 for the far-hue hint in 3B against 0.958 $[0.939, 0.974]$ held out, and the inflation is 0.002–0.035 across the four models and three hint conditions (largest for LLaVA, $0.958 \to 0.923$; Gemma-3-4B inflates by 0.010–0.019, $0.982 \to 0.964$ for the plausible hint). Scene overlap therefore helped the earlier numbers estimate, but it does not account for the ranking: the held-out probe still separates the conflict arms far above chance and within 0.04 of its fitting-scene value.

The regularisation is fixed at $C = 0.1$ for every reported probe number. Sweeping $C \in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 0.1\}$ under the held-out protocol moves the conflict AUROC by at most 0.020 (LLaVA, plausible hint, 0.906–0.926), 0.012 in 3B and 0.016 in 7B; for the far-hue hint the spread is at most 0.016 in every model. No choice of C changes the ordering—conflict below no hint, correct at or near ceiling—or the size of the output-signal collapse, which does not depend on the probe at all. We report the fixed $C = 0.1$ rather than the grid-selected value so that the headline number is not chosen on the evaluation set.

C Per-colour breakdown

Table 7: Verified samples by gold colour. Counts are the 261 manifest items; the absent arm uses the category's most frequent colour, so it is not restricted to this set.

colour	blue	white	black	green	red	yellow
n	88	79	45	25	15	9

Blue and white dominate the manifest. A per-colour AUROC is not estimable at useful precision for red and yellow ($n = 15$ and 9), and we do not report one. The colour distribution is a property of COCO val2017 under our geometric filter, not a design choice; the absent hint uses each category's own colour distribution.

D Controls for text priors and strict absence

A gray image with the same prompts and dimensions yields probe AUROC 0.849/0.981/0.930 under no/conflicting/correct hints, and p(`unknown`) 0.700/0.325/0.433. Text alone carries some category-to-colour prior, enough to reach 0.85 on the no-hint pair. But the direction learned from gray images transfers across

hints at only 0.540/0.603, whereas the real-image direction transfers at 0.958/0.990 on held-out scenes (Table 2). The real-image direction is shared across prompts; the text-prior direction is not.

Category-matched pairs hold the question fixed, which the main design does not: there the present and absent arms query different categories, so a present–absent separation can be carried by the question and the asserted colour as well as by the image. In the control, each present image is paired with a different image in which the same category is annotated absent, the question is identical, and the same colour is asserted in both arms (the category's most frequent val colour, which is never the present image's gold), so neither the question nor the hint can predict the label. There are 260 pairs over 30 categories. Table 8 reports the result.

Table 8: Category-matched control. The question and the asserted colour are identical in both arms of a pair, so neither can predict the label; cross-validation groups by category. "Frozen" fits on no-hint states of training categories and scores the conflict condition; "refit" retrains inside it. Margins are frozen probe minus $p(\texttt{unknown})$, category-grouped bootstrap 95% CIs. Geometry is the length ratio, rotation and along-axis projection of the present–absent mean difference.

model	no hint	frozen	refit	$p(\texttt{unknown})$	frozen − $p(\texttt{unknown})$	geometry
3B	0.983	0.867	0.975	0.817	$+0.050$ $[0.028, 0.077]$	0.565 / 40° / 0.435
7B	0.983	0.823	0.978	0.820	$+0.002$ $[-0.025, 0.045]$	0.394 / 46° / 0.274
LLaVA	0.994	0.855	0.983	0.870	-0.015 $[-0.062, 0.025]$	0.586 / 66° / 0.240
Gemma	0.947	0.887	0.933	0.896	-0.009 $[-0.032, 0.011]$	0.546 / 24° / 0.500
Phi	0.961	0.592	0.950	0.446	$+0.145$ $[0.063, 0.232]$	0.547 / 82° / 0.081

The output collapse, the in-condition decodability and the geometry all survive. $p(\texttt{unknown})$ falls from 0.983 to 0.817 in 3B, from 0.975 to 0.820 in 7B, from 0.989 to 0.870 in LLaVA, from 0.930 to 0.896 in Gemma and from 0.891 to 0.446 in Phi, and maximum-probability AUROC from 0.952 to 0.699, from 0.967 to 0.768, from 0.949 to 0.510, from 0.904 to 0.783 and from 0.903 to 0.357; refitting recovers 0.975, 0.978, 0.983, 0.933 and 0.950; and the present–absent mean difference shortens to 0.565, 0.394, 0.586, 0.546 and 0.547 of its no-hint length and turns 39.6°, 45.9°, 65.9°, 23.8° and 81.5°, against 0.523/0.384/0.602 and 38.5°/47.4°/70.7° for the Qwen-family plausible hint on the main pair (Table 3). The Phi case also separates the two sources of dilution: on the main pair the vector keeps only 0.052 of its length, but within a fixed question it keeps 0.547 while still turning 82°, so much of the main-pair shortening is carried by the change of queried category, not by the hint alone. What does not survive is the size of the frozen transfer. With the question and the hint held constant across the label, the no-hint probe scores 0.867, 0.823 and 0.855, and its margin over $p(\texttt{unknown})$ is $+0.050$ $[0.028, 0.077]$, $+0.002$ $[-0.025, 0.045]$ and -0.015 $[-0.062, 0.025]$, against $+0.123$, $+0.142$ and $+0.142$ on the main pair; in 7B and LLaVA the margin is indistinguishable from zero, and in LLaVA the point estimate is negative. Gemma follows: frozen 0.887, margin -0.009 $[-0.032, 0.011]$. Phi is the exception, with frozen 0.592 and margin $+0.145$ $[0.063, 0.232]$, but this is a comparison against an own signal at 0.446, below chance, and the probe itself is far from the 0.950 refit; it is the least bad score, not a good one. A substantial part of the main design's frozen transfer is therefore carried by the change of queried category and asserted colour between the arms, not by the evidence alone. The refit probe is not exposed to this confound, because it is fitted where question and hint are constant, and the geometry is computed within the pair; both remain. Grouping folds by image instead of category removes donor-image reuse (157 images serve as both a present image and a negative) and leaves the frozen and refit values within 0.01 (0.876/0.984 in 3B, 0.833/0.980 in 7B, 0.869/0.989 in LLaVA).

At the same 25% answered coverage (130 of 520 items), the frozen probe's fabrication rate on the conflict pair is 0.185 (3B), 0.154 (7B), 0.146 (LLaVA), 0.131 (Gemma) and 0.438 (Phi), against 0.254/0.154/0.054/0.115/0.446 for $p(\texttt{unknown})$ and 0.346/0.169/0.600/0.254/0.592 for maximum probability. In 3B the gate still beats the model's own signal within-question, but by 0.069 instead of 0.115 on the main pair; in 7B the probe and $p(\texttt{unknown})$ coincide; in LLaVA and Gemma the probe is worse than $p(\texttt{unknown})$; and in Phi it is better by only 0.008 at this coverage while ranking better (0.592 against 0.446). The 7B point is weakly informative because it answers only 26% of matched conflict items, so 25% of all items selects 95% of its answers; LLaVA answers 90%, so its point is informative and unfavourable, and Phi's 25% point selects 34% of items. The main-design gate advantage is therefore partly an arm-diagnostic advantage, and we report both. Excluding the 17 of 260 negatives that the presence probe judges present above 0.5 leaves this unchanged: $p(\texttt{unknown})$ is 0.992 without hints and 0.851 under conflict, and maximum probability 0.692 and 0.473. The main absent arm is checked the same way: per-item presence probabilities on all 261 absent images give mean $p(\text{yes}) = 0.046$, and 7 items are judged present above 0.5. On the remaining 254, refusal under the conflicting hint is 0.531 (against 0.521 on all items), refusal with no hint is 0.921, hint adoption is 0.457, and the evidence probe reaches AUROC 0.999; the strict-absence restriction therefore does not change the main result. For LLaVA the presence check is much

weaker (mean p(yes) = 0.403, 73 items above 0.5), so its yes/no readout is not a reliable absence detector; on the 183 items it judges absent, conflict refusal moves only from 0.088 to 0.104, adoption is 0.891 and the probe remains at 1.000. For LLaVA, "absent" therefore rests on the annotation rather than on the probe.

A model without an abstention behaviour. We also ran the main battery on SmolVLM-Instruct (2B; Idefics3 vision encoder, SmolLM2 language model) as a non-Qwen backbone. It never emits `unknown`: refusal is 0 in all six cells, and it adopts the stated colour on 0.985 (hint a) and 0.989 (hint b) of absent items, so there is no behavioural gate to study and we do not use it as a replication. Its final state nonetheless separates present from absent items, at probe AUROC 0.989 with no hint and 0.993 under the conflicting hint (p(known) 0.921 and 0.598; maximum probability 0.877 and 0.257). A decodable evidence axis can therefore exist in a model that never abstains. Because the conflict-pair probe for this model could also pick up hint–image mismatch rather than evidence, we treat this as a boundary observation about the axis, not as evidence about the gate.

Scene fit of the absent category. The absent category is sampled without regard to whether it fits the scene, so abstention could track implausibility rather than missing evidence. For each item we score the absent category by its COCO train co-occurrence with the categories annotated in that image, $s = \text{mean}_c\, P(\text{absent} \mid c)/P(\text{absent})$, and split at the median (Table 9). The score predicts no-hint refusal in every model, so the absent arm does carry a scene-fit component: the better-fitting half refuses less by 0.116 $[0.047, 0.185]$ in 3B, 0.069 $[0.023, 0.115]$ in 7B and 0.132 $[0.039, 0.224]$ in LLaVA. Category frequency is a rival explanation, since frequent categories are also refused less in 7B (0.073 $[0.025, 0.128]$) and LLaVA (0.100 $[0.008, 0.195]$), though not detectably in 3B (0.062 $[-0.014, 0.133]$). The two scores are only weakly related (Spearman $\rho = 0.23$). Fitting no-hint refusal on both jointly with a scene bootstrap, scene fit remains negative in all three models, at -0.81 $[-1.26, -0.37]$ (3B), -0.68 $[-1.24, -0.29]$ (7B) and -0.44 $[-0.78, -0.16]$ (LLaVA) per standard deviation, while frequency is negative only in 7B (-0.45 $[-0.98, -0.14]$) and indistinguishable from zero in 3B (-0.29 $[-0.76, +0.11]$) and LLaVA (-0.01 $[-0.34, +0.37]$). The scene-fit difference is also present within each frequency half (Table 10). It does not account for the evidence signal. Within each half p(`unknown`) still separates present from absent images, at AUROC 0.988–0.998 in 3B, 0.976–0.991 in 7B and 0.986–0.995 in LLaVA, and the probe at 0.992–0.999, 0.989–1.000 and 0.993–0.997. Under the conflicting hint the stratum difference is not significant in 7B (-0.047 $[-0.124, +0.037]$) or LLaVA ($+0.024$ $[-0.045, +0.093]$) and remains in 3B (-0.165 $[-0.287, -0.050]$), where hint adoption is 0.157 $[0.042, 0.280]$ higher in the better-fitting half. The same conclusion therefore holds in both strata: plausibility moves the level of the gate, not the ranking of evidence.

Table 9: Scene-fit stratification of the absent arm ($n = 261$ per model, median split). "lo"/"hi" are the lower and upper halves of $s = \text{mean}_c\, P(\text{absent} \mid c)/P(\text{absent})$, computed from COCO train co-occurrence with the categories annotated in the image. "AUROC" is p(`unknown`) separating present from absent under no hint; "probe" is the internal probe on the same pairs. Refusal and adoption use hint a.

model	half	n	refuse, none	refuse, a	adopt, a	AUROC / probe
3B	lo	131	0.962	0.603	0.389	0.998 / 0.999
	hi	130	0.846	0.438	0.546	0.988 / 0.992
7B	lo	131	0.992	0.901	0.099	0.991 / 1.000
	hi	130	0.923	0.854	0.146	0.976 / 0.989
LLaVA	lo	131	0.885	0.076	0.916	0.995 / 0.997
	hi	130	0.754	0.100	0.892	0.986 / 0.993

Table 10: No-hint refusal in the 2×2 of scene fit s (rows) and category frequency (columns), both median-split; cell sizes are 80, 51, 56 and 74 items from top to bottom. Scene fit lowers refusal in every frequency stratum.

cell	3B	7B	LLaVA
s lo, freq lo	0.963	1.000	0.887
s lo, freq hi	0.961	0.980	0.882
s hi, freq lo	0.893	0.982	0.839
s hi, freq hi	0.811	0.878	0.689

Prompt format. The main battery offers an explicit `unknown` option and a bracketed hint, which could overstate abstention or hint uptake. We rerun the absent arm three ways in all five models (Table 11). Removing the `unknown` option makes abstention impossible by construction (refusal is 0 in every condition) and raises adoption of the plausible hint from 0.467 to 0.935 in 3B, from 0.123 to 0.977 in 7B, from 0.904 to 0.992 in LLaVA, from 0.579 to 0.789 in Gemma and from 0.663 to 0.985 in Phi-3.5. Phrasing the hint as a sentence rather than a bracket leaves refusal under conflict at 0.958/0.989 in 3B/7B, raises it from 0.088 to 0.559 in LLaVA, from 0.402 to 0.851 in Gemma and from 0.322 to 0.877 in Phi-3.5, while reducing hint adoption to 0.038/0.004/0.333/0.038/0.088; accuracy under the wrong hint rises from 0.383 to 0.594 in 3B, from 0.249 to 0.498 in 7B, from 0.360 to 0.548 in LLaVA, from 0.241 to 0.559 in Gemma and from 0.080 to 0.276 in Phi-3.5. The conflict effect is thus tied to a prefixed assertion and to the availability of an abstention token; a conversational hint does not reproduce it.

Table 11: Prompt-format robustness ($n = 261$ absent items per model). "std" is the main battery; "no `unknown`" removes the `unknown` option from the system prompt; "sentence" phrases the hint as a sentence. Hint adoption for absent b is measured against the far-hue colour actually stated.

model	format	refuse, none	refuse, hint a	refuse, hint b	adopt a	adopt b
3B	std	0.904	0.521	0.563	0.467	0.433
	no `unknown`	0.000	0.000	0.000	0.935	0.908
	sentence	0.931	0.958	0.981	0.038	0.011
7B	std	0.958	0.877	0.958	0.123	0.042
	no `unknown`	0.000	0.000	0.000	0.977	0.973
	sentence	0.958	0.989	0.992	0.004	0.000
LLaVA	std	0.820	0.088	0.188	0.904	0.762
	no `unknown`	0.000	0.000	0.000	0.992	0.946
	sentence	0.793	0.559	0.736	0.333	0.088
Gemma	std	0.843	0.402	0.475	0.579	0.502
	no `unknown`	0.000	0.000	0.000	0.789	0.640
	sentence	0.812	0.851	0.870	0.038	0.027
Phi-3.5	std	0.897	0.322	0.425	0.663	0.567
	no `unknown`	0.004	0.000	0.000	0.985	0.950
	sentence	0.931	0.877	0.931	0.088	0.046

Input resolution. LLaVA runs at a different input resolution from the Qwen models, so we rerun the 3B battery at 262,144 and 589,824 pixels. At 262,144, absent refusal is 0.908 with no hint and 0.556 under the plausible hint, hint adoption is 0.441, accuracy under the wrong hint is 0.433, and the frozen probe reaches AUROC 0.997; at 589,824 the values are 0.904, 0.521, 0.467, 0.383 and 1.000, matching the native run. Resolution moves the level by at most 0.05 and does not produce the conflict effect.

E External validation details

VizWiz is used for naturally unanswerable questions and VQA-v2 for colour questions with real answer distributions. The VizWiz run keeps 462 items (212 answerable, 250 unanswerable) and the same three prompt conditions; the hint is the most frequent colour of the queried category, as in the plausible condition. The VQA-v2 run uses 300 colour questions over 277 images, with the conflicting hint taken from the far-hue mapping. Both runs use the same 3B model and the probe frozen after fitting on the main no-hint pair.

Table 12: VizWiz (Qwen2.5-VL-3B). Refusal and accuracy by condition, and AUROC over answerable versus unanswerable questions. In-domain means refitting the probe inside VizWiz; frozen means transferring the probe fitted on the 261-image set.

condition	ans. refuse	ans. acc	unans. refuse	probe (in-dom.)	probe (frozen)	p(`unknown`)
no hint	0.099	0.580	0.704	0.909	0.827	0.878
correct hint	0.033	0.816	—	—	—	—
conflicting hint	0.193	0.451	0.636	0.803	0.720	0.793

On VizWiz the asymmetry is present but weaker than in the constructed design: the hint lowers unanswerable refusal by 0.068, whereas the constructed conflict lowers it by about 0.36, and it raises answerable refusal by 0.094. Maximum probability is close to chance on both pairs (0.437/0.367), while $p(\texttt{unknown})$ is strong (0.878/0.793). The frozen probe adds nothing over $p(\texttt{unknown})$ on this data (-0.051 $[-0.078, -0.023]$ without a hint, -0.074 $[-0.119, -0.031]$ with one). Naturally unanswerable questions are failed for heterogeneous reasons, and the hint does not target the refusal gate, so the regime in which the probe wins is not reached. At 50% coverage, the risk of an incorrect answer is 0.457 for the probe, 0.449 for $p(\texttt{unknown})$ and 0.655 for maximum probability; none of the three is usable, because half of these questions have no valid answer.

Table 13: VQA-v2 colour questions (300 items, 277 images, Qwen2.5-VL-3B). "Score" is the frozen probe score; "> 0" is the fraction it classifies as evidence-present. AUROC is for answering correctly on answerable questions. Risk is $1 -$ accuracy at the stated coverage.

condition	accuracy	$p(\texttt{unknown})$	max prob.	score	> 0	AUROC
no hint	0.850	0.024	0.865	+6.93	0.940	0.710
correct hint	0.983	0.021	0.971	+6.70	0.983	—
conflicting hint	0.613	0.115	0.722	+2.10	0.780	0.489

VQA-v2 reproduces the direction of the effect without the constructed absent arm: the score falls by 4.83 points and the present-classification rate falls from 0.940 to 0.780, while accuracy falls from 0.850 to 0.613. The score is correlated across conditions ($r = 0.54$), so the shift is graded by the item rather than uniform. The ordering survives in the sense that the probe still separates correct from incorrect answers without a hint (0.710), but under conflict it is at chance (0.489), and maximum probability is the better correctness predictor in both conditions (0.831/0.649). At 50% coverage on no-hint answers, maximum probability has risk 0.007 against 0.087 for the probe. The probe is an evidence detector, not a correctness predictor, and on real answer distributions the two come apart.

F Second stimulus set: PASCAL VOC 2012

The main set is built on COCO. To check that the construction rather than a COCO peculiarity produces the pattern, we rebuilt the six cells on PASCAL VOC 2012 segmentation [9]: 180 images from the val split (long side at most 1024 px), each contributing one present item and one annotation-absent item, with present instances passing the same component, box-size, fill and colour-purity gates as the main set and absent categories being VOC classes with no annotated pixel in the image, hinted with that class's most frequent colour in the split. Labels come from the automatic gates only, with no contact-sheet verification, and the same Qwen2.5-VL-3B checkpoint, prompts and readout are used.

Behaviour again moves both arms in opposite directions. Without a hint the model refuses 0.894 of absent items and names the present item's colour correctly on 0.422; the plausible hint lowers absent refusal to 0.644 and is adopted on 0.339 (far hue: 0.683 and 0.311), while on present items the wrong hint raises refusal from 0.011 to 0.400 and lowers gold accuracy from 0.422 to 0.217. The correct hint raises accuracy to 0.839.

The probe transfers. Fitted on the no-hint pair with scene-grouped folds it reaches AUROC 0.994 [0.987, 0.999], and refitted inside the conflict pair 0.998 [0.996, 1.000]. Fitted on training scenes' no-hint pairs and applied unchanged to held-out scenes it reaches 0.983 [0.971, 0.992] under the correct hint, 0.925 [0.895, 0.955] under the plausible hint and 0.950 [0.921, 0.975] under the far-hue hint.

The geometry replicates. The present–absent mean difference shortens to 0.61, 0.41 and 0.52 of its no-hint length and turns 28° [26°, 31°], 35° [33°, 38°] and 33° [31°, 37°] from the no-hint axis; per-scene coherence falls from 0.61 to 0.43, 0.31 and 0.39, and the two hinted colours coincide at $\cos = 0.89$ [0.84, 0.92]. The effect is not a COCO artefact.

Patching localises the gate on this stimulus set as well. Pasting the image's own no-hint absent state back into the conflict pass recovers -24% at layer 16 (44% depth), 29% at layer 24 (67%), 84% at layer 30 (83%) and 100% at layer 34 (94%); a shuffled absent source gives -28%, 24%, 98% and 102%. Recovery is again late, steep and category-level rather than sample-specific.

The law's form transfers but its gain does not, and the difference is diagnostic. The within-class spread contracts to 0.88, 0.82 and 0.79 of its no-hint value, against 0.71 on the main set. Because the d' ratio is the gap-to-spread ratio, the through-origin slope fitted here is 1.20, and the main-set slope 1.425 over-predicts the three cells by 0.088–0.175 (RMSE 0.123). Per cell the inverse spread contractions are 1.14, 1.21 and 1.27, so the

fitted slope again tracks 1/spread contraction, as it does on the main set ($1/0.71 = 1.41$ against a fitted 1.42). The linear relation is preserved; the constant multiplying it is a property of how much noise the hint removes in a given stimulus set and model, not a universal gain.

Signal detection and the gate follow the main set. Sensitivity falls from $d' = 3.44$ to 0.67 under conflict (Δ [−3.52, −2.35]) while the criterion moves from +0.48 to −0.09 (Δ [−0.89, −0.34]); the correct hint leaves $d' = 2.61$. At 25% answered coverage the frozen probe's fabrication rate is 0.144 [0.078, 0.211], against 0.367 [0.272, 0.444] for p(`unknown`) and 0.422 [0.356, 0.517] for maximum probability (paired difference [−0.300, −0.133]); at 50% coverage, where the model answers 0.478 of conflict items, all three tie at 0.372 (paired difference [−0.020, 0]). As on the main set, the gate helps at low coverage and not at high.

G Second backbone outside the Qwen family: Phi-3.5-vision

Gemma-3-4B was our first non-Qwen point, and its output signal does not collapse. To test whether the collapse is tied to the Qwen2/2.5 readouts or to one training recipe, we ran the full battery on Phi-3.5-vision-instruct (a Phi-3-mini language model with a CLIP ViT-L/14-336 encoder), loading its native remote code with eager attention because its config pins flash-attention-2, and using its native prompt format. All six cells and the final-token state (3072-d) were collected for the same 261 scenes; labels, prompts, readout and analysis are unchanged. It qualifies as an abstaining backbone: it refuses 0.897 of absent items without a hint.

Conflict moves both arms in opposite directions and, in this model, past each other. Absent refusal falls from 0.897 to 0.322 (hint `a`) and 0.425 (hint `b`), with adoption 0.663 and 0.567; on present items the wrong hint raises refusal from 0.119 to 0.548 and lowers gold accuracy from 0.625 to 0.080, while the correct hint raises it to 0.858. Present refusal therefore exceeds absent refusal under conflict, the reverse of the no-hint ordering. The own-signal detector follows: p(`unknown`) AUROC falls from 0.939 [0.918, 0.958] to 0.385 [0.336, 0.433] and 0.458 [0.408, 0.506], below chance in both hinted cells, and maximum probability is near chance throughout (0.442/0.492/0.492). On the same items the internal probe reaches 0.992 [0.987, 0.997] with no hint and 0.998 [0.996, 1.000] refitted inside the conflict pair.

The geometry is the extreme case of the pattern. The present–absent mean difference shortens to 0.525 [0.470, 0.575] under the correct hint and to 0.052 [0.006, 0.113] and 0.149 [0.096, 0.212] under the plausible and far-hue hints, and turns 52.8° [49.4°, 56.8°], 85.9° [80.9°, 89.5°] and 74.4° [67.4°, 80.2°] from the no-hint axis. Per-scene coherence falls from 0.388 to 0.234, 0.027 and 0.070, and the two hinted colours still point the same way (cos = 0.720 [0.646, 0.790]). The near-orthogonal rotation is why the frozen probe falls to 0.954 [0.936, 0.971] under the correct hint and to 0.685 [0.640, 0.731] / 0.675 [0.633, 0.718] under the two conflicting hints, the weakest transfer here, even though the same states are decodable at 0.998 when the readout is refitted. This is the sharpest form of the dissociation: the evidence is present and linearly decodable, but the no-hint direction has almost no projection onto it. The linear law keeps its form but not its gain: applying the four-model slope 1.425 out of sample over-predicts Phi's three cells by 0.024, 0.075 and 0.145 (RMSE 0.096), and Phi's own through-origin slope is 1.13, below every fitted model's gain, so the constant remains model-specific as reported above.

Signal detection is a full reversal rather than a criterion shift: d' goes from 2.43 [2.18, 2.71] to −0.44 [−0.63, −0.26] (Δ [−3.19, −2.58]) and the criterion from −0.04 [−0.17, 0.10] to +0.10 [+0.01, +0.21]; the correct hint leaves $d' = 2.57$. The gate still works where the own signal has reversed: at 25% answered coverage the frozen probe's fabrication rate is 0.285 [0.208, 0.377], against 0.562 [0.477, 0.631] for p(`unknown`) and 0.662 [0.577, 0.723] for maximum probability (paired difference [−0.354, −0.192]), and at 50% coverage it is 0.556 [0.498, 0.609] against 0.586 [0.544, 0.632] and 0.636 [0.594, 0.678] (paired difference [−0.073, −0.004]). The advantage is large at low coverage and marginal at high, as in the main models. The category-matched control and the conformal scan, run after an earlier draft, do not change this. Within a fixed question the frozen probe reaches 0.592 [0.530, 0.657] against 0.446 for p(`unknown`) and 0.950 for the refit, and the gate's fabrication rate at 25% answered coverage is 0.438 against 0.446 (probe) and 0.592 (maximum probability): the ranking margin over a below-chance own signal does not translate into a meaningful operating-point advantage. Conformal certification is infeasible at both $\alpha = 5\%$ and 10% in all 20 splits (minimum UCB 0.305). The prompt-format controls reproduce the scope conditions: removing `unknown` raises adoption from 0.663 to 0.985, sentence phrasing raises conflict refusal from 0.322 to 0.877 and cuts adoption to 0.088, and wrong-hint accuracy rises from 0.080 to 0.276 (Table 11). Patching localises the gate as in the other backbones: the image's own no-hint absent state recovers 31% at layer 14 (44% depth), 97% at layer 22 (69%), 102% at layer 28 (88%) and 101% at layer 30 (94%); the present-question state drives refusal below base at the deep layers (−0.27 and −0.33), and shuffled

absent sources recover as much as the own state (1.07/1.05). Its role is scope: the geometry replicates in a third language-model family and a third vision encoder, the output collapse is not specific to Qwen, and the gate is written late, while Gemma remains the counterexample that separates the representation from the readout.

H Site and direction controls in the two non-Qwen backbones

Table 14 reports the head- and layer-site injections that accompany the single-site results in Section 4.5. All settings use the conflicting pass, the direction fitted on the model's own no-hint states, and the same readouts. Head rows use $\pm 0.50\, w$; layer rows use ± 0.25 of that layer's median last-token norm.

Table 14: Site controls in Gemma-3-4B and Phi-3.5-vision ($n = 261$, conflicting pass). Entries are changes from the unsteered base in absent-object refusal for the negative and positive dose. "Present move" is the largest absolute change in present-item gold accuracy or hint adoption over the two signs: in Gemma large values at L16 and L24 mark collateral disruption, whereas in Phi the direction moves the present arm coherently with the absent arm. The last column is absent-item hint adoption. Head rows inject at the unembedding input, layer rows at the output of the stated decoder layer.

model	site	Δ absent refusal (−)	Δ absent refusal (+)	present move	absent hint (−/+)
Gemma	head	+0.056	−0.109	0.019	−0.061 / +0.105
	L8 (24%)	−0.146	−0.103	0.088	+0.142 / +0.111
	L16 (47%)	+0.280	−0.339	0.674	−0.494 / −0.111
	L24 (71%)	+0.010	−0.092	0.375	−0.262 / +0.065
	L30 (88%)	+0.088	−0.151	0.023	−0.151 / +0.096
Phi	head	+0.435	−0.330	0.195	−0.437 / +0.322
	L14 (44%)	+0.180	−0.193	0.107	−0.176 / +0.190
	L22 (69%)	+0.057	−0.111	0.046	−0.057 / +0.113
	L28 (88%)	+0.036	−0.136	0.042	−0.038 / +0.142
	L30 (94%)	+0.103	−0.128	0.050	−0.109 / +0.130

Gemma's head is movable but not direction-specific. The fitted direction moves absent refusal monotonically ($+0.056$ $[0.029, 0.082]$ at -0.50 and -0.109 $[-0.140, -0.082]$ at $+0.50$), but less than random unit directions orthogonal to it at the same norm (mean absolute effect 0.076 against 0.042 for w, ratio 0.55), and ablating it is null (-0.002 $[-0.006, 0.000]$) although it removes $5.9\times$ the energy of a random projection. In the layers, L16 moves the gate in both directions but also collapses present-item gold accuracy and hint adoption (present move 0.674), L24's negative direction is similarly destructive (0.375), and L8 lowers refusal at both signs. L30 (88% depth) is the only clean site: absent refusal -0.151 $[-0.184, -0.121]$ at -0.25 and $+0.088$ $[0.059, 0.117]$ at $+0.25$, absent hint $+0.096/-0.151$, and present gold and hint unchanged (present move 0.023). Patching recovers 90% at L24 and 101% at L30, so the gate is written late in Gemma as well; what fails is the claim that w is a privileged readout direction.

Phi's direction is privileged. At the head it is $3.0\times$ the mean random effect on absent refusal (0.220 against 0.073) and $12.9\times$ for ablation, and it moves present refusal and present gold coherently with absent refusal (-0.349 and $+0.195$ at $+0.50$), as an answer/refuse axis should. Layer injection is already large at L14 (44% depth; absent refusal -0.193 $[-0.232, -0.159]$ at -0.25 and $+0.180$ $[0.148, 0.215]$ at $+0.25$, present gold $+0.100/-0.038$) and weakens at L22–L30, while patching recovers only 31% at L14 and 97% at L22. The axis is therefore available to the readout before the conflict-induced loss is localised: injection establishes sufficiency of the direction, not the site where the conflict acts. Neither model has a random-direction control at layer sites, so the layer profiles are descriptive rather than direction-specific.

I Per-layer probe

A probe refitted independently at each layer (grouped by scene) reaches AUROC 0.500 at the embedding output, 0.867 at layer 4, 0.982 at layer 8, and ≥ 0.990 from layer 16 onward; transfer from the no-hint pair to the conflicting pair has a dip at layer 8 (0.803) and recovers to 0.965–0.995 deeper. Refitting, however, gives each layer its own estimator, so this curve measures decodability rather than a shared axis. We therefore also fit the estimator once on the final layer (per fold) and apply its weights unchanged to every layer, with per-layer standardisation only. That shared readout is at chance through position 18 of the stored stack (embedding plus

36 layers; AUROC ≤ 0.53, rising to 0.58 and 0.64 at positions 19 and 21), then jumps to 0.86 at position 22, 0.97 at position 24 and 0.99 from position 25 onward. The rise is not monotone (0.67 at position 23), so we locate it to the last quarter rather than a single layer. Evidence is thus linearly decodable early, but in a layer-specific basis; the final-layer evidence axis used by steering and patching is assembled late. The two analyses share a target but not an estimator, and we do not map probe AUROC at layer L one-to-one onto recovery at layer L.

J Full dose–response

Table 15: Steering the no-hint direction during the conflicting pass (Qwen2.5-VL-3B, $n = 261$). "True" and "hint" are argmax rates on present items; "hint" on absent items is the rate of naming the hinted colour, averaged over hints a and b.

setting	present refusal	present true	absent refusal	absent hint
no hint, unsteered	0.011	0.762	0.904	—
conflict, base	0.088	0.383	0.542	0.450
$-0.50\,w$	0.931	0.008	0.941	0.057
$-0.25\,w$	0.759	0.073	0.874	0.125
$+0.25\,w$	0.000	0.479	0.065	0.845
$+0.50\,w$	0.000	0.479	0.000	0.665

K SDT across models and task forms

Criterion shifts under conflict are $+0.16$ $[-0.18, +0.39]$ (3B), -0.54 $[-0.73, -0.35]$ (7B) and $+1.00$ $[+0.66, +1.33]$ (LLaVA); d' falls in all three. Under a correct hint, d' changes are $+0.31$, $+0.50$ and $+0.32$.

Table 16: Signal detection (Qwen2.5-VL-3B). d' and c are computed from paired present/absent refusal rates. Separation is the fixed probe's mean gap in pooled within-arm SD units. Slopes use z from the no-hint state (fixed) or the condition's own state (own), with the same direction in both.

condition	d'	c	separation	slope, fixed	slope, own
no hint	3.51	0.46	5.06	-0.719	-0.719
conflicting hint	1.45	0.62	2.60	-0.171	-0.559
correct hint	3.83	0.62	4.17	-0.680	-0.641

Table 17: Behavioural check of the dilution law. g is the along-axis gap ratio on the same pair; d'_{out} is computed from the logit of $p(\texttt{unknown})$ and d'_{rate} from argmax refusal rates, each relative to the no-hint cell. The ratio d'_{out}/g is the behavioural-to-geometric gain. The no-hint axis is re-estimated inside every bootstrap draw.

model	hint	g	d'_{out} ratio	d'_{rate} ratio	d'_{out}/g	$p(\texttt{unknown})$ AUROC
3B	a	0.409	0.354	0.397	0.87	0.834
	b	0.433	0.424	0.427	0.98	0.859
7B	a	0.259	0.375	0.341	1.45	0.797
	b	0.307	0.431	0.526	1.40	0.840
LLaVA	a	0.199	0.280	0.345	1.41	0.781
	b	0.234	0.427	0.479	1.83	0.879
Gemma	a	0.466	0.626	0.797	1.34	0.890
	b	0.496	0.724	0.867	1.46	0.922

L Construct and predictive validity

Two checks separate what the probe measures from what it does not. Construct validity asks whether the present–absent separation is an artefact of answer correctness: restricting the present arm to items the model answered *wrong* leaves the separation essentially unchanged, and within-present correct-vs-wrong is far weaker. The probe tracks evidence presence, not answerability.

Table 18: Out-of-sample checks for the two dilution laws (`pilot/law_validation.py`). LOCO fits on $n-1$ cells and predicts the held-out cell; LOMO fits on the other models and predicts all cells of the held-out model; the shared rows fit one through-origin slope to every cell against one slope per model. The probe-law bootstrap re-estimates the no-hint axis, the gap and the d' cells inside each of 500 scene resamples.

check	probe law	behavioural law
cells	12	8
LOCO MAE (training-mean / unit-slope baseline)	0.056 (0.202 / 0.179)	0.100 (0.125 / 0.121)
LOMO MAE, 3B / 7B / LLaVA / Gemma	0.12 / 0.02 / 0.02 / 0.12	0.21 / 0.02 / 0.07 / 0.25
RMSE, shared slope vs. per-model slope	0.060 vs. 0.011	0.097 vs. 0.029
through-origin slope [95% CI]	1.425 [1.37, 1.47]	1.27 (slope fit 0.97, $R^2 = 0.57$)

Table 19: Task form with the same images and absent categories (Qwen2.5-VL-3B). The gate opens in all three; the output readout collapses mainly for colour.

task	condition	refusal	adopts hint	probe	p(`unknown`)
colour	present, conflict	0.088	0.448	0.997	0.834
	absent, no hint	0.904	—	0.995	0.993
	absent, hint	0.521	0.467	—	—
side	present, conflict	0.000	0.268	0.991	0.966
	absent, no hint	0.621	—	0.993	0.989
	absent, hint	0.111	0.743	—	—
count	present, conflict	0.870	0.115	0.995	0.946
	absent, no hint	1.000	—	0.996	0.986
	absent, hint	0.996	0.004	—	—

Predictive validity asks whether the score predicts individual conflict behaviours. It predicts absent-item refusal in 3B, 7B and Gemma; in Gemma it also predicts whether a correct present answer survives and whether the hint is adopted. It does not predict adoption in 3B/7B, predicts nothing in LLaVA, where refusal is near zero and adoption near ceiling, and in Phi-3.5 the conflict scores point the wrong way (AUROC 0.392 for absent refusal, 0.376 for present gold and 0.440 for adoption), consistent with its weak frozen transfer. Construct validity holds in Phi: the no-hint probe separates present from absent at 0.992 even on present items the model answers wrongly (0.987) and is far weaker within the present arm (0.680). The gate therefore works on average without being a calibrated per-item predictor of credulity: the score measures evidence availability, not the strength of the hint override.

M Deployment details

Scores are scene-cross-fitted for all five models: the probe is fitted on the no-hint pairs of four scene folds and applied to the conflict pairs of the held-out fold. All deployment numbers use the plausible hint `a` and pair `sufficient|wrong` with `absent|a`, giving 522 items. Fabrication is the fraction of answered items that are absent-object items; coverage is the fraction of all items the system answers, and each scorer's threshold is chosen to reach the stated answered coverage. The 3B refit gate reaches 0.100 at 50% answered coverage; the cross-fitted score reaches 0.115. The 12-dimensional logit readout reaches 0.142 at 50% selection coverage. Conformal calibration uses half the scenes, a Clopper–Pearson upper bound at 90% confidence over a 50-point threshold scan, family-wise correction over the scan, and evaluation on the other half; the answered subset is used in both numerator and denominator.

N Compute

All runs used one NVIDIA A10 with 24 GB. The main 3B run (six conditions per image, hidden states saved) takes about 30 minutes; the 7B and LLaVA runs take about an hour each; the Gemma-3-4B run takes about six minutes at a single 512-pixel resize; the full suite, including steering, patching and external validation, fits in a few GPU-hours.

Table 20: Construct validity: no-hint probe AUROC over paired items (95% scene bootstrap).

model	present vs absent	present answered-correct	present answered-wrong	within present, correct vs wrong
3B	0.995 [0.989, 0.999]	0.996 [0.989, 1.000]	0.991 [0.980, 1.000]	0.619 [0.538, 0.697]
7B	0.996 [0.993, 0.999]	0.998 [0.995, 1.000]	0.991 [0.981, 0.999]	0.572 [0.492, 0.652]
LLaVA	0.994 [0.989, 0.998]	0.995 [0.988, 0.999]	0.994 [0.985, 1.000]	0.602 [0.532, 0.671]
Gemma	0.985 [0.974, 0.994]	0.985 [0.969, 0.996]	0.985 [0.969, 0.995]	0.586 [0.515, 0.656]
Phi-3.5	0.992 [0.986, 0.996]	0.996 [0.991, 0.999]	0.987 [0.972, 0.997]	0.680 [0.613, 0.747]

Table 21: Predictive validity: out-of-fold conflict score against conflict behaviours (AUROC, 95% scene bootstrap).

model	absent refuse	present gold	present adopt
3B	0.644 [0.581, 0.708]	0.549 [0.478, 0.623]	0.448 [0.379, 0.520]
7B	0.695 [0.587, 0.794]	0.567 [0.481, 0.653]	0.503 [0.405, 0.601]
LLaVA	0.454 [0.346, 0.566]	0.529 [0.450, 0.606]	0.477 [0.404, 0.552]
Gemma	0.750 [0.689, 0.807]	0.676 [0.596, 0.753]	0.334 [0.260, 0.414]